
Politecnico di Torino

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# Software Design and Architecture

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# Chapter 1

# Introduction to Software Design

## 1.1 Complexity

Complexity is anything related to the structure of a software system that makes it hard to understand and modify the system.

$$C = \sum_p c_p t_p$$

Where $C$ is determined by all software parts $p$, $c_p$ is the complexity of part $p$, and $t_p$ is the fraction of time spent on part $p$.

### 1.1.1 Syntax complexity

- Change amplification: change require operating in different places
- Cognitive load: how mutch information to learn to understand and perform a task. Some times is better to write longer code to reduce the complexity and so the cognitive load.
- Unknown unknowns: there are partes of the system that we don't know where they are or even that they exist.

### 1.1.2 Cause of complexity

**Dependency**: When we need to understand or modify a part of the code, we need to understand or modify other parts of the code. Try to reduce the number of dependencies and the complexity of the dependencies.

**Obscuiry**: alla the important information shoul be clearly visible.

### 1.1.3 Strategical vs. Tactical Programming

- **Tactical programming**: focus on having code that works as fast as possible. "Move fast and break things".
- **Strategical programming**: focus on long term maintainability and scalability. "Working code is not enough".

## 1.2 Modules

To improve and eliminate dependency and complexity, we can use modules. A module is a part of the code that has a well defined interface and a well defined implementation.

- **Interface**: what we need to know to use the module. Formal part, possible checked by the compiler. Represents a cost.
- **Implementation**: how the module works. Ideall much more complex than the interface. Represents a benefit.

### 1.2.1 Abstraction

What we tipically do is to perform an abstraction. Abstraction means selective removal of information. Is a simplified view of the something that omits "unimportant" information. Omitting information can lead to eads to unknowns and keeping unimportant information can leads to cognitive overload.

#### Deep modules and shallow interfaces

**Deep modules**: modules with a simple interface and a complex implementation. They are good because they reduce the cognitive load and the change amplification. They are bad because they can lead to unknown unknowns.
**Shallow interfaces**: modules with a complex interface and a simple implementation. They are good because they reduce the unknown unknowns. They are bad because they can lead to cognitive overload and change amplification.

### 1.2.2 Information Hiding

Information hiding is the principle of hiding the implementation details of a module from the users of the module. Example sort function: we want to sort a list of numbers, we don't care about how the sorting algorithm used.

### 1.2.3 Information Leakage

Is the opposite of information hiding. Occurs when the implementation details of a module is reflected in multiple modules with consequences of creating dependencies.

### 1.2.4 Information Hiding Red Flags

- **temporal decomposition**: execution order is reflected in the code structure. Operations that happen in different times are in different classes.
- **overexposure**: API forces knowledge about rarely used features to use common ones. E.g. `BufferedReader r = new BufferedRea`
- **Too many classes**: code is divide into too many shallow classes.

### 1.2.5 Method separation

Better together or Better apart?

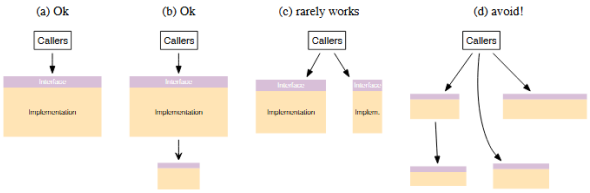
- **Bring together**: if information is shared, if we simplify the interface, if we eliminate duplication.
- **Separate**: if the code is general purpose or specific purpose.

#### Method separation red flags

If we have a lot of repeded code and/or we mixed specific and general code, means we have put together code that should be separated.

### 1.2.6 Do one simple thing and complexity

Each method should do one thing and do it completely




## 1.3 Obscurity

Obscurity is the principle of making the important information clearly visible and the unimportant information hidden. Obscurity is a great source of complexity. Obscure code is hard to Understand and creates more Bugs.

The solution to obscurity is writing obvious code, this means with a quick glance we can understand what the code does. We can achieve this by using abstraction layers.

### 1.3.1 Abstraction layers

Software systems are composed in layers. Higher layers use the facilities provided by lower layers.

Abstraction simplifies understanding and reduces obscurity.

In a well-designed system, each layer provides a different abstraction from the layers above and below it.

#### Abstraction Layers Red Flags

Some red flags that indicate that we have a problem with abstraction layers are:

- **Pass-Through methods**: is a method that does nothing but call another method.
- **Pass-Through variables**: is a variable that is passed through over and over down a long chain of method calls.

### 1.3.2 Writing Comments

Another method to reduce obscurity is writing comments. If users need to read a method code to use it, there is no abstraction.

Comments should describe things that are not obvious in the code:

- **Pick conventions**: write comments like other persons in the project
- **Don't repeat code**: avoid redundant comments that simply restate what the code already makes clear
- **Higher level comments enhance intuition**
- **Delayed comments are bad comments**
- **Use comments as design tool**

### 1.3.3 Naming

Toghether with comments, naming is a great tool to reduce obscurity. Bad names cause bugs. Name should create a image in mind of the reader about the nature of the thing being named. Names should be precise and consistent.

## 1.4 Errors and Anomalies

Exceptions add complexity because they create dependencies between the code and if they are easy to throw they are hard to handle.

A piece of code might throw exceptions in different ways:

- Caller may provide bad arguments or configuration
- Invoked method may not able to complete the request
- In distributed systems network packets may be lost or delay or peers may communicate in unxpected ways
- The code may detect bugs or internal inconsistencies or situations that are not prepared to handle

### 1.4.1 Define Errors out of Existence

What are the solutions to errors? Define errors out of existence. This means, not ignore the error but to redefine the behavior of a method so that no error is required.

E.g. `String.substring(int beginIndex, int endIndex)` it throws different exceptions Some in only specific cases. This is completly avoidable and unnecessary.

### 1.4.2 Limit exceptions

There are some methods to limit the number of exceptions that a method can throw:

- Handle exceptions at the point where they are thrown, if possible
- Exception aggregation means handle different exceptions with same handler
- Just crash
- Mask exceptions avoiding unexpected exceptions

## 1.5 Dependency Dimensions

### 1.5.1 Structural (Code-Level) Dependencies

- **Implementation Dependency**: some piece of code depends on a concrete class instead of an abstraction
- **Construction Dependency**: you depend on how a collaboration, another class, is created.
- **Compile-Time Dependency**: the module knows something at compile time

### 1.5.2 Runtime / Distribution Dependencies

Often linked to distributed architectures.

- **Spatial Dependency**: A component depends on the location of another component. E.g. hard-coded URLs
- **Temporal Dependency**: A component depends on another being available at the same time. E.g. remote method calls or servers
- **Synchronization Dependency**: A component depends on another to complete work before continuing. E.g. synchronous request/response

### 1.5.3 Data-Level Dependencies

- **Data Dependency**: One component depends on another's internal data model. E.g. shared database
- **Schema Dependency**: Multiple services depend on the same database schema or representation structure. E.g. DB schema, XML/Json schema

### 1.5.4 Behavioral Dependencies

- **Control Flow Dependency**: One component dictates another's execution path. E.g. Centralized process managers
- **Transactional Dependency**: E.g. Components rely on atomic distributed consistency.
- **Timing Assumption Dependency**: Assumptions about response time or ordering. E.g. "Service must respond in <50ms" or Event ordering assumptions

# Chapter 2

# Design Patterns

Initially proposed by Christopher Alexander in the context of architecture.

A reusable solution
to a known problem
in a well defined context

The context is a *design* situation giving rise to a (design) problem.
The problem is a set of forces repeatedly arising in the context. [1]
The solution is a proven resolution of the problem. Configuration to balance forces. Structure with components and relationships and run-time behaviour.

**Example of Social Pattern:** You are in queue to the supermarket gastronomic section. The problem is hard to spot who arrived first. The solution is to use a ticket machine.

## 2.1 Types of Software Patterns

- **Architectural patterns** - high level structures of software systems. E.g., layered architecture, client-server, microservices.
- **Design patterns** - solutions to common design problems. E.g., Singleton, Observer, Factory Method.
- **Idioms** - language-specific patterns. E.g., RAII in C++, list comprehensions in Python.

### 2.1.1 Architectural pattern

Normally the architectural pattern works with big components and their interactions and is needed to describe the responsibilities of these components. Expresses a fundamental structural organization schema for software systems and defines the rules and guidelines for organizing the relationships between the components.

**Example of architectural pattern:** Several programs that are used in sequence read from input and write sequentially to output.
The problem is there are a lot of intermediate files used for communication between programs.
The solution is to adopt pipes and filters architecture feeding with the result of the previous one.

### 2.1.2 Design patterns

Provides a scheme for refining components of a software system or their relationships and describes a commonly recurring structure of communicating components.

**Example of design pattern:** A class library providing few functionalities contains a lot of classes.
The problem is the user is exposed to the internal complexity of the library.
The solution is to create a new façade class that interacts with the user and hide all the details.

### 2.1.3 Idioms

It is a low-level pattern specific to a programming language. Describes how to implement particular aspects of components or the relationships between them. Leverages the features of a programming language.

[1] Any relevant aspect of the problem E.g., requirements, constraints, desirable properties

**Example of idioms:** An attribute is constant and should be globally available to many classes.
The problem is opening access would allow unauthorized modifications and the attribute is repeated in every object.
The solution is to make public static final in Java language.

## 2.2 Pattern language

Pattern do not exist in isolation in fact can happen that two or more patterns are applied together, used to implement part of another pattern and introduce a problem solved by another
They are called Pattern Languages or pattern systems.

Collection of patterns together with guidelines for:

- Implementation
- Combination
- Practical use

Should:

- Count enough patterns
- Describe patterns uniformly
- Present relationships

## 2.3 Idioms

**X-ability**

Clonability, serializability, comparability, etc. some specific behaviour dealing with many heterogeneous classes. The solution is to derive from a class or interface that defines the required methods.

**Example:** In Java all methods extends the root class *Object* and provides the methods *toString()*, *equals()*, *hashCode()*, etc. that can be overridden to provide specific behaviour.

### 2.3.1 Strong typing vs. Duck Typing

*"If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck."*
**Strong typed language:** You must define interfaces and classes to be able to call a specific method. Compile time type checking.
**Dynamic/duck typed language:** You just add a method to a class and you can call it. No compile time type checking.

### 2.3.2 Property

A property, more general than an attribute, is a named characteristic of an object typically accessed through well-defined operations called accessors, getters, and setters forming the public interface of the object and it is not necessarily an attribute of the object (e.g. the age can be derived from the date of birth).

## 2.4 Design Patterns

Description of communicating objects and classes that are customized to solve a general design problem in a particular context.
A design pattern names, abstracts, and identifies the key aspects of a common design structure that make it useful for creating a reusable object- oriented design.

| | | **Purpose** | | |
|---|---|---|---|---|
| | | Creational | Structural | Behavioral |
| **Scope** | Class | 1 | 1 | 2 |
| | Object | 4 | 6 | 10 |

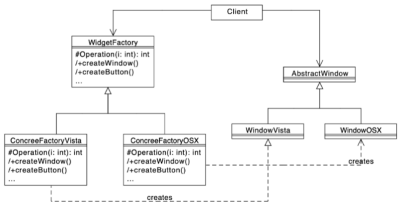
**Pattern selection** How to use patterns? First of all you need to know the problem you are trying to solve, knowing the patterns and then select the one that best fits your needs.

### 2.4.1 Creational patterns

- **Class**: Factory Method
- **Object**: Abstract Factory, Builder, Prototype, Singleton

### Abstract Factory

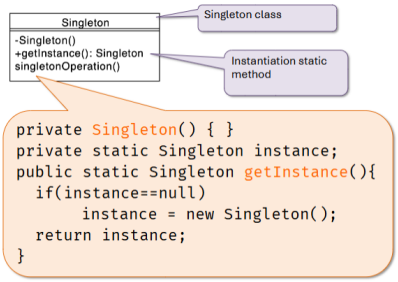
A family of related classes can have different implementation details.
The client should not know anything about which variant they are using / creating.




**Example:** GUI where you have different components (buttons, text fields, etc.) and the same GUI can be applied to different platforms such as Windows, MacOS, Linux, etc.

### Singleton

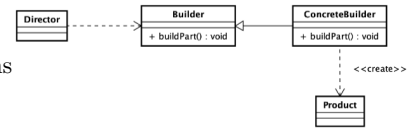
A class represents a concept that requires a single instance.
The client should be able to access, in approper way, the unique instance.




> **Note:** If we talk about of an object designer pattern, in python you can just create a global variable and it will be a singleton.

### Builder

An object of a complex class has to be created.
The creation entails complex interaction with the object and could be different variations of the object.



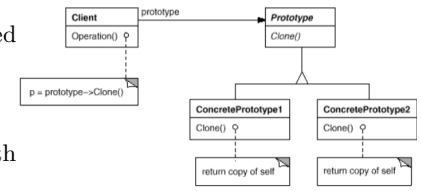

**Example:** $10.40\,kg \cdot m^2 \cdot s^{-3}$

```
//Usual non-fluent builder
Measure power = new Measure(10.4);
power.addUnit("kg", 1);
power.addUnit("m", 2);
power.addUnit("s", -3);
power.setPrecision(2);
```

```
//Fluent builder
Measure power = new Measure.Builder(10.4)
    .is("kg").by("m")
    .squared().by("s").to(-3)
    .withPrecision(2).build();
```

### Prototype

Several object can be created out of a palette of similar classes.
The class hierarchy can become quite complex and classes in hierarchy are differentiated by minor parameters (that could be defined through constructor arguments).
A solution could be start with a prototype classe and clone it to create new objects.




**Example:** A graphical editor where you have very similar shapes (e.g. square with different radius, colors, etc.).

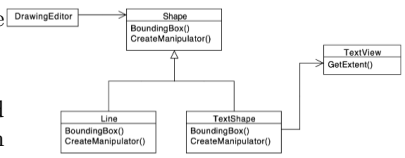
## 2.4.2 Structural patterns

How organize classes and objects to compose larger structures.

- **Class**: Adapter
- **Object**: Bridge, Composite, Decorator, Facade, Flyweight, Proxy

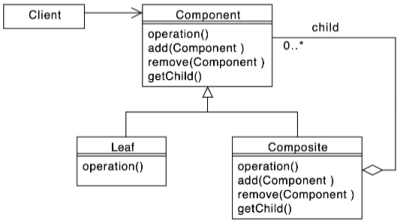
### Adapter

A class provides the features required by another class but its interface is not the one expected by the client.
The integration of the provider class should be possible without modifying it. The source code could be not available or is already used as it is somewhere else.




**Example:** In Java, keyboard events are handled by KeyListener. Listener class need to implement the interface and implement all the methods. If you are interested only in one of the methods, you can use the KeyAdapter class that is empty and you can decide which method to override, and the Listener extends KeyAdapter instead of KeyListener.

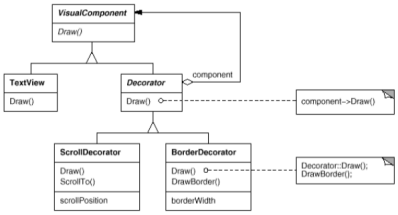
**Composite**




You need to represent part-whole hierarchies of objects.
Clients can be very complex and they require to adapt the difference between all the elements in the hierarchy.

**Example:** In Java UI components are organized in a tree structure. You can have a panel that contains other panels and buttons, etc.

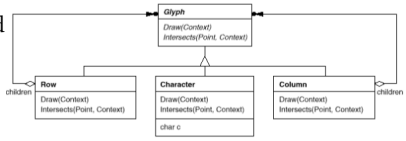
**Decorator**




You have some objects of classes that have specific behaviour and you want to add the same behaviour plus someting else dynamically.
The solution can be extend the class but if that behaviour is something you would to apply to many classes, you need to extend all of leaf classes. Using a decorator you can modyfy the behaviour without creating an explosion of subclasses for every combination of features

**Example:** In Java we have JScrollPane can be used to provide a scroll pane to any component

**Flyweight**




You need need to create a very large number of similar objects, and memory usage and performance become concerns.

**Example:** In Java we can pool different objects togheter. The Integer class in Java.

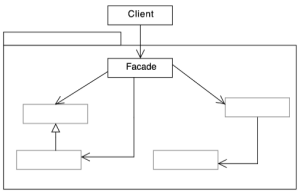
**Proxy**

You need to controll access to an object.

We need to provide a placeholder that is able to mediate between the client needs and all the requirements you want to controll before accessing the final object without changing interface or class.

- **Remote proxy**: provides a local representative for an object in a different address space.
- **Virtual proxy**: creates expensive objects on demand. E.g. slow loading resource - images in a web page.
- **Protection proxy**: controls access to the original object. E.g. when objects should have different access rights.
- **Smart reference**: is a replacement for a bare pointer that performs additional actions when an object is accessed. E.g. counting the number of references, checking permissions, etc.

**Facade**




Hide the details and the complexity of a general class by providing a simpler interface.
The client should not know anything about the subsystem.

### 2.4.3 Behavioral patterns

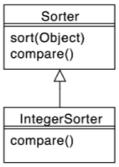
Behavioral patterns allow to implement specific algorithms using objects and classes to provide control and flexibility.

- **Class**: Interpreter, Template Method
- **Object**: Chain of Responsibility, Command, Iterator, Mediator, Memento, Observer, State, Strategy, Visitor

The basic mechanism of behavioral patterns is enapsultion variation. Instead of having a lot of if-else you have a structure at object level that allows you to define different alternatives.

### 2.4.4 Template Method

An algorithm/behavior has a stable core and several variation at given points. You have to implement/maintain several almost identical pieces of code.




**Example:** The compare in Java. You have to implement the compare method that solves integer or string comparison.

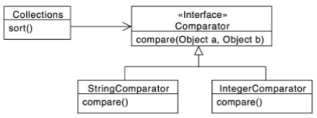
### 2.4.5 Strategy Pattern

Many classes or algorithm has a stable core and several behavioral variations, several different implementations are needed.
Every variation of an algorithm can be embedded in an object and passed as a parameter to the algorithm.

**Example:** Comparator in Java.

**Consequences**:

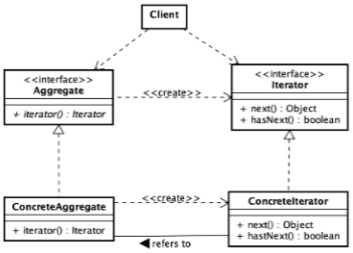
+ Avoid conditional statements
+ Algorithms may be organized in families
+ Choice of implementations
+ Run-time binding
- Clients must be aware of different strategies
- Communication overhead
- Increased number of objects




### 2.4.6 Iterator pattern

You have a collection of objects and you want to iterate on it allowing multiple concurrent iterators.

**Example:** In Java there is the Iterable interface and Iterator.

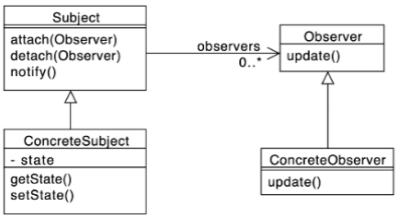



### 2.4.7 Observer pattern

The change in one object may influence one or more other objects. This can cause high coupling between objects and the number and type of objects to be notified may not be known in advance.

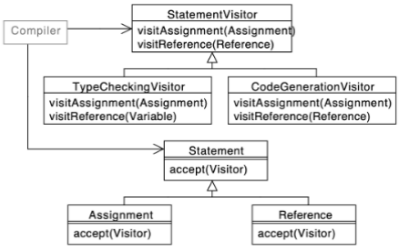
**Example:** In Java there is the Observable interface and Observer interface. This is very common in UI programming.
Consequences:

+ Abstract coupling between Subject and Observer
+ Support for broadcast communication
- Unanticipated updates




### 2.4.8 Visitor

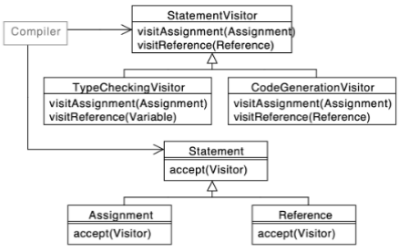
You have an object structure that contains many classes with different interfaces and you want to perform several different and unrelated operations on these objects.
The operations depend on the concrete classes and you want to avoid putting many different methods in each of these classes.



## 2.5 Inheritance vs. composition

Reuse can be achieved via:

- **Inheritance**: a new class is created by inheriting from an existing class and adding new features or modifying existing ones. Clients can invoke directly inherited methods.
- **Composition**: The class has a link or instance to a class you want to reuse and just forwards the calls to it.




**Note:** The Observable pattern is deprecated. The PropertyChangeSupport is more flexible.

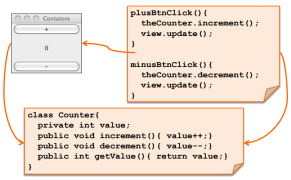
## 2.6 Abstract Syntax Tree

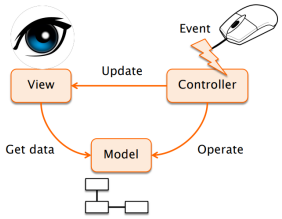
The AST is a tree that represent a syntax of formal language. We can use the composite pattern to representationt he arithmetic expressions.

And using the visitor pattern or the interpreter pattern to evaluate the expression.

## 2.7 Model View Controller

**Context**: Interactive applications with flexible HCI.
**Problem**: The same information is presented in different ways/windows, Windows must present consistent data, Should be possible to add/change presentations of information




### 2.7.1 MVC Principles

When building a GUI we must consider two main aspects:

- Layout (**View**): how to place the graphical elements to achieve a give visual aspect
- Events (**Controller**): which behavior associate to elements' events
- Application logic (**Model**): must remain, as far as possible, separate from user interface.

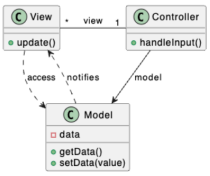



Figure 2.1: MVC

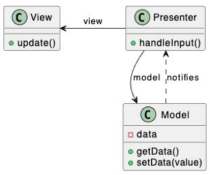
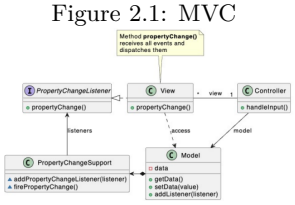
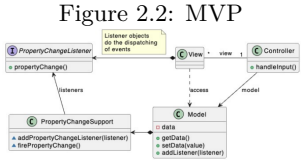



Figure 2.2: MVP





### 2.7.2 Execution flow

There is no predefined order of execution. Operation are performed in response to external events (e.g. mouse click). Event handling is serialized. To execute operations in parallel threads must be used. Method main in GUIs has the only goal of instantiating the graphical elements

## 2.8 Summary

Architectural patterns deal with overall system structure. They provide a unique metaphor for the system (e.g. pipe and filters). They address specific domains (e.g. distribution or interaction) and system evolvability

# Chapter 3

# What is software architecture ?

The architecture of a software system is the shape given to that system by those who build it. The form of that shape is in the division of that system into components, the arrangement of those components, and the ways in which those components communicate with each other. The purpose of that shape is to facilitate the development, deployment, operation, and maintenance of the software system contained within it.

## 3.1 Goals of an architecture

The software must change to adapt to software regulation, user feedback.

We have some pattern that make methods and classes easy to be changed, but now we want something behond the methods.

**The goals of an architecture are** :

- keep the system valuable for its users over time
- minimize the lifetime cost of the system
- maximize programmer productivity

**This can be achieved by making the system** :

- easy to understand
- easy to develop
- easy to maintain
- easy to deploy
- easy to change

In Java we have the package, that is a module, and is a logical grouping of classes.

**Design Patterns:** helps to build methods and classes inside the module in a way that they are easy to be changed.

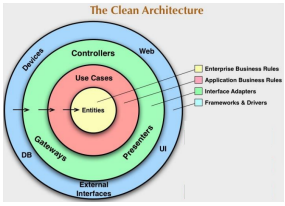
**Design Principles:** define the modules can collaborate together and how they are shaped.

## 3.2 The clear architectures

The **Outer circles**: mechanisms, they change often.

The **Inner circles**: policies, subject to less frequent changes.

Source code dependencies point only inward, toward higher-level policies (Dependency Rule)

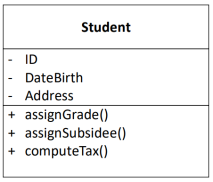



### 3.2.1 Entities

An Entity is an object within a software system that contains a set of critical business rules operating on Critical Business Data.

The interface of the Entity consists of the functions that implement the Critical Business Rules to operate on the Critical Business Data. They contain the most general and highlevel rules and neither they are often changed nor they are usually affected by lower-level changes.

**Example:** The system is the university portal and an important entity is the student, that has some critical business rules like "the student can enroll to the master degree only if they have completed the required prerequisites and then assign new ID".
Is the most stable part of the system and we want to preserve as much as possible from changes occurring to the other elements.



## 3.3 Use cases

To applay the Businness Roles to the entities we need to use the use cases.
A use case is a description of the software behavior and interactions (with people or other software) in a specific case.
It specifies:

- the input to be provided by the user (or other systems)
- the output to be returned
- The processing steps involved.

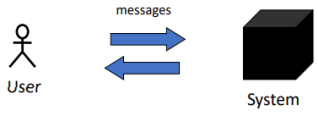
to determine the flow of data to and from the entities.
Software at this level change whenever a change to any element of a use case is required

**Use case in pratice**

It represents a **functional requirement**.

- It makes clear what you expect from a system ("what?")
- It hides the behaviour of the system ("how?")
- It is a sequence of actions (with variants) that produce a result observable by an actor

The system is seen as a black-box. Use cases express the interaction (exchange of messages) between the entities that interact with the system itself (called actors).




**Example - Specify annual tax for a student** Input: student ID
Output: annual tax amount

Main scenario:

1. Validate ID provided by user
2. Retrieve economic level
3. Compute due past taxes
4. Compute new taxes
5. Get average grades for the last academic year
6. If average grade $> 26$, apply 20% discount
7. Ask operator for confirmation: if yes change student else reset

## 3.4 Interface adapters

Their goal is to shape data in a way that it is convenient to higher levels or other systems or artifacts interacting with:
The same is done in the opposite direction. Examples of code in this layer: Presenters for GUI, Controller for DB
Code inward to this level know anything about details (e.g., SQL)

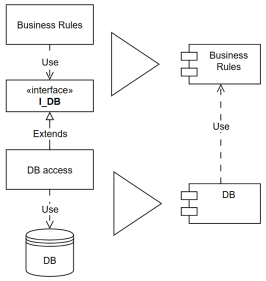
## 3.5 Frameworks and drivers

In order to achieve the separation we have seen so far, we need to separate low level details from the high-level details. Layers allow independent development, deployment, operation, and maintenance.

The details are the frameworks, the database, the drivers, the web server, etc., and usually we do not write mutch code here.

## 3.6 Separation of policy from details

Decoupling of policy from the details and defer decisions on details for as long as possible (MongoDB or SQL, etc.)
DB component knows about use case rules component, but NOT viceversa.
Low level components know everything of the higher-level components, but NOT viceversa.
The Database component contains the code that translates the calls made by the BusinessRules into the query language of the database. This translation code knows about the BusinessRules.
Business rules can be adapted to any kind of DB.



# Chapter 4

# From design to architecture: Design Principles

Design principles means follow a clean architecture and a clean architecture means that the architecture can easily evolve, to be changed.

## 4.1 The SOLID design principles

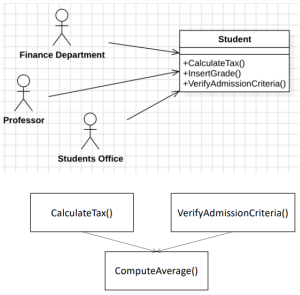
The SOLID principles suggest software architects on how to arrange functions and data into groupings (e.g., classes), and how these groupings are arranged.

The goal of the SOLID principles is to create mid-level software structures (modules) that:

- Are easy to understand
- Allow changes in the most "comfortable" way
- Are the basis of reusable components

### 4.1.1 Single Responsibility Principle - SRP

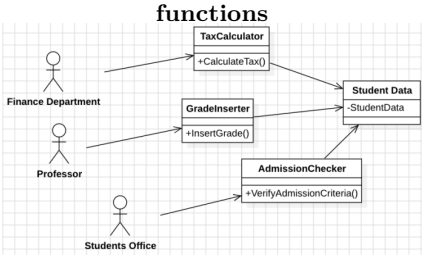
A module should have one, and only one, reason to change. It means that a module should be responsible to **one**, and only one, **actor**.
This is helpful to separate code used by different actors and for a better maintenance.



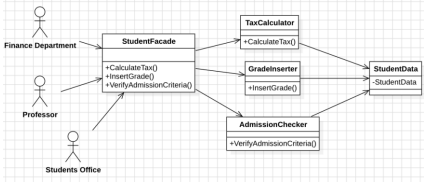

**Example** *CalculateTax()* is specified by the Finance Department.
*InsertGrade()* is specified by Students Office and used by Professor
*VerifyAdmissionCriteria()* is specified and used by Students Office
*CalculateTax()* and *VerifyAdmission()* share a common algorithm for calculating the average grade.

Changes in *ComputeAverage()* affect two actors: errors may arise and also changes to Class Student due to different needs of actors may lead to collisions and merges.

**Solution 1: Separate data from functions**




**Solution 2: Use the Façade pattern** You increase cupling and complexity, its a trade off, but the separation is splitted.

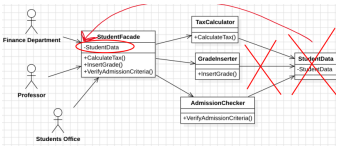



**Note:** You increase cupling and complexity but the separation is splitted.

**Solution 2: variation**

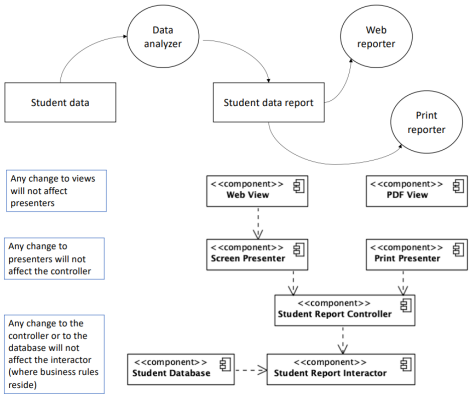
## 4.2 Open-Closed Principle - OCP

A software artifact should be **open for extension but closed for modification**. Using design patterns to make new funcitonality in a way that is addictive and not to modify the existing structure.


Finance Department
Professor
Students Office
StudentFacade
-StudentData
+CalculateTax()
+InsertGrade()
+VerifyAdmissionCriteria()
TaxCalculator
+CalculateTax()
GradeInserter
+InsertGrade()
AdmissionChecker
+VerifyAdmissionCriteria()
StudentData
-StudentData

The behavior of a software artifact ought to be extendible, without having to modify that artifact.

The aim of OCP is to make the system easy to extend without requiring too many changes to the system.

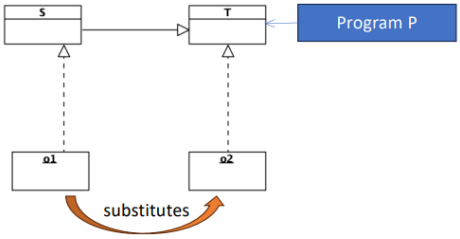



Using the open-closed principle, the software architect divides the system into components and arranges these components in a dependency hierarchy that protects higher-level components from changes in lower-level components.

**Example** A system displays summary information about a student's career at the university on a dashboard: Subjects (those passed with grades), Taxes paid and due, Current semester schedule, Next exams, Important notices, Stakeholders ask for a printed report with only information on subjects and taxes, and a different formatting.

Generating the report involves two separate responsibilities:
1. the calculation of the reported data
2 .the presentation of that data into a web- and printer-friendly form.

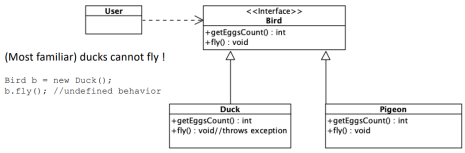
### 4.2.1 Liskov substitution principle - LSP

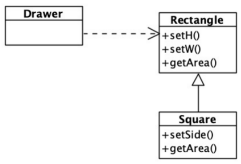



If for each object o1 of type S there is an object o2 of type T such that for all programs P defined in terms of T, the behavior of P is unchanged when o1 is substituted for o2 then **S is a subtype of T**.

If we substrute o2 vitw o1 and the behavior of p does not change means S is derived class of T.

This principle is about subtyping, not only for inheritance or extension but also for the logic, the behavior of the class.
Derived type must fully support the substitution of their base types: functions that use pointers or references to base classes must be able to use objects of the derived without knowing it. This is related to the substitution property.
A subtype can replace its supertype only if it behaves the same way. This idea, called **behavioral subtyping**, means the subtype doesn't just have the same methods as the supertype — it also follows the same rules for how those methods should work. That way, anything the client expects from the supertype will also hold true for the subtype.



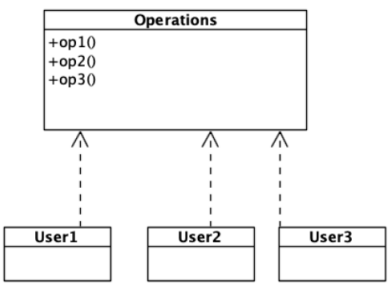




```
Rectangle r = ...;

r.setW(5); r.setH(2);
assert(r.area() == 10);
```

Depending on how r is defined, the assertion can fail because you not use the *setH()* and *setW()* methods of the rectangle but you use the *setSide()* method of the square.

**How to refactor it?** Using the Interface segregation principle - ISP

### 4.2.2 Interface segregation principle - ISP




Clients should not be forced to depend on operations and/or interfaces they do not use. Better to make fine grained interfaces that are client specific.
Similart to SRP we have different actors, represented by different classes. They are only interested in a subset of the operations, so we can split the interface in U1 use only operation1 and U2 use only operation2 and U3 use only operation3

In a statically typed language (e.g., Java), any change to *op1()* will force recompilation and redeployment of User2 and User3.

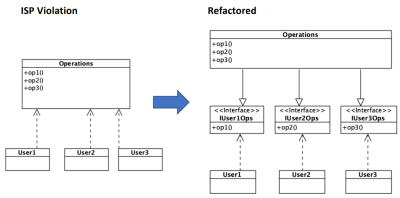
Is different from the SRP because we need the classes but we need to modularize better.

Inserting interfaces the classes use the interfaces instead the class and the interface implement the class.

If we change operation1 we impact also user2 and user3, if we change the interface we impact only the corrisponded user.

We also package the interface and the user toghether and this not impact the module.

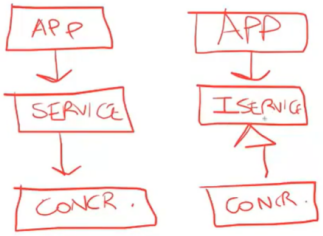
Operations are segregated into interfaces.
In a statically typed language (e.g., Java), a change to *op1()* will NOT force recompilation and redeployment of User2 and User3.




Remove useless operations from modules, otherwise they will force useless redeployments

### 4.2.3 Dependency inversion principle - DIP

You should always have dependencies that lower modules depend on higher modules, not the opposite.
**High-level modules should not depend on anything from low-level modules. Stable abstractions: abstractions should not depend on details, so that changes to details will not affect them.**
**Idealistic tension**, in practice each system has at least a concrete component: the main function
We can make an interface:



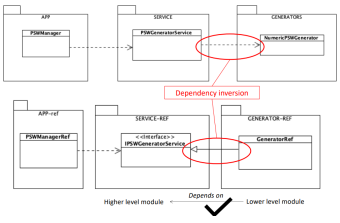

**Example** In Java (or other statically typed languages), this means that use, import and include statements should refer only to source modules containing interfaces, abstract classes or any other type of abstraction.

In dynamically typed languages, source code dependencies should not refer to modules in which functions being called are implemented.

Also we can package the app and the interface together.

### 4.2.4 DIP: practical implications

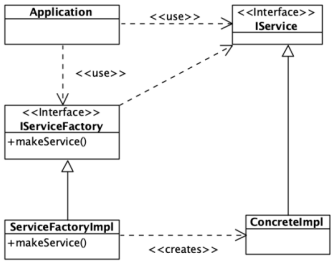
Dependencies on concretions are tolerable when it comes to operating systems, platform facilities and other similar components whose changes are tightly controlled and not frequent.
**Abstract Factories are helpful** to adhere to the DIP by minimizing the number of dependencies on concrete implementations and keep them separate from the rest of the system
In general, volatility should be shifted towards implementation details without changing interfaces.




- Don't refer to volatile concrete classes
- Don't refer from volatile concrete classes
- Don't override concreted functions
- Never mentions the name of anything concrete and volatile

**Using Factory Pattern**

To create an instance of ConcreteImpl, the Application calls *makeService()* from IServiceFactory interface, which is implemented by ServiceFactoryImpl class.
ConcreteImpl is created and returned as a IService.




**Separation abstract and concrete levels:** All source code dependencies point toward the abstract level. The flow of control crosses the separation line in the opposite direction: this is the sense of "dependency inversion".

**DIP violation** DIP violations cannot be entirely removed. Some dependencies on concrete implementations are necessary. They must be minimized and kept separate

## 4.3 How to compile and see the UML

`mvn -f .\Path\to\pom_file\pom.xml clean compile site`
Then Java Project > Target > site > index.html

# Chapter 5

# Components

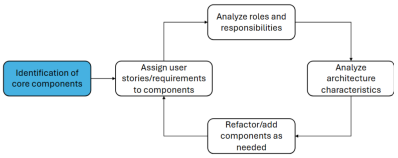
The logical components are the building blocks of the system. They usually manifest through a namespace or a directory structure, where leaf nodes containing source code represent the logical components.

**Logical architecture**: the system's logical components and how they interact each other.

**Physical architecture**: includes also physical artifacts, defining architectural

## 5.1 Identify components

**Example:** SPG = Solidarity Purchasing Group, Solidarity towards producers valuing (small vs large, local and seasonal vs global and specific vs mass). Solidarity among clients: Not a normal shop but a group of people buying together, sharing the principles to select producers.
This is a software application for **Clients**, **Farmers** and **Shop employees**.




The identification of components is fundamental activity, necessary either for drafting architecture from scratch or reverse-engineer it. It is an iterative refinement process and three approaches can be used for initial identification:

- **Best guess**: based on the system core functionalities
- **Workflow**: Based on the "happy path"
- **Actor/action**: Based on major actions a user can perform

#### SPG requirements

A neighborhood association manages a Solidarity Purchasing Group (SPG). Every Monday, a farm (rotating among a pool of reference suppliers) submits to the SPG a list of available products, their quantities, and their prices.

Once this list is provided, the data is reviewed and confirmed by the SPG employee.

SPG members receive an email with the list of available products and can place their orders either via the website or by visiting the association's headquarters.

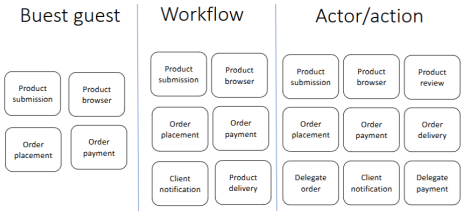
When ordering, SPG members specify the types of vegetables they want and the quantities (expressed in kilos, packs, or boxes depending on the type of fruit or vegetable ordered). If the order is placed at the association's headquarters, the member pays immediately.

At the end of the week, the farm delivers the products to the association's headquarters. Shop employees record the delivery, and members receive a notification when their ordered goods arrive. At this point, members can go to the association to collect the package (paying at the same time, if not done at the time of ordering).

| **Macro functionality** | **Component** |
|---|---|
| Insert products | Product submission |
| View products | Product browser |
| Order products | Order placement |
| Buy products | Order Payment |

| Step | Component |
|---|---|
| 1. Farmer submits available products, employee reviews | Product submission |
| 2. SPG members/clients receive list of products | Client notification |
| 3. Client browse the catalog of product | Product browser |
| 4. Client places an order | Order placement |
| 4a. Employee places an order for the client | Order placement |
| 5. Client pays the order online | Order Payment |
| 5a. Employee submit order payment | Order Payment |
| 6. Employees register delivery of goods | Product delivery |
| 7. Clients are updated on goods' delivery | Client notification |
| 8. Clients retrieve goods | Product delivery |

| Actor | Action | Component |
|---|---|---|
| Client | Search for products | Product browser |
| | View details about products | Product browser |
| | Place order | Order placement |
| | Pay order | Order payment |
| Farmer | Insert products availabilities | Product submission |
| Shop employee | Review products availabilities | Product review |
| | Insert client order | Delegate order |
| | Insert client payment | Delegate payment |
| | Register product delivery | Order delivery |
| System ! | Send email notifications | Email Notification |




**Workflow approach**

**Actor/action**

### 5.1.1 Putting components together

- **Tactic level - design**: Components principles: cohesion, coupling
- **Strategic level - architecture**: Component topology, technical partitioning vs domain partitioning, and Physical architecture, Monolithic vs Distributed architecture.
- **And more**: Deployment, communication style and data topology

## 5.2 Component principles

### 5.2.1 Component cohesion

How to put classes inside a component?

Cohesion is the degree to which the classes within a component belong together.

**High cohesion** means that the classes within a component are closely related in terms of functionality and purpose, while **low cohesion** indicates that the classes are more unrelated and serve different purposes.

It is suggested to follow principle:

- The **Reuse/Release Equivalence Principle - REP**: The granule of reuse is the granule of release.
- The **Common Closure Principle - CCP**: Classes that change together are packaged together.
- The **Common Reuse Principle - CRP**: Classes that are used together are packaged together.

#### Reuse/Release Equivalence Principle - REP

To support reuse, classes and modules that make up a component must: belong to a coherent group and be released together.

The release process shall produce appropriate notifications and release documentation, so that users can decide whether to integrate new releases.

#### Common Closure Principle - CCP

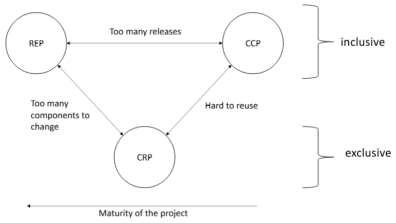
Group into components classes that change together, i.e. for the same reason and at the same time.

Separate into different components classes that have different changes processes and motivation.

> **Note:** There is a relationship between the Common Closure Principle (CCP). It is similar to SRP and it addresses the "closure" aspect of OCP

**Common Reuse Principle - CRP**

It help deciding which classes/module to place into the same components. In such components, usually classes have many dependencies on each other
Complementary perspective:




- users of a component should not depend on unnecessary features;
- if a class isn't closely connected to another class, it shouldn't be in the same component.

Similar to interface segregation design principle (ISP).

> **Note:** Similar to interface segregation design principle (ISP)

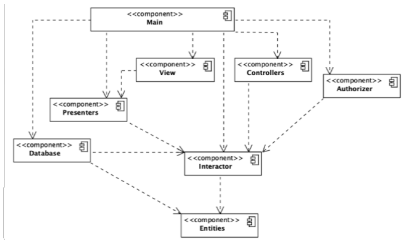
### 5.2.2 Component coupling

- The Acyclic Dependencies Principle (ADP): The dependency graph of packages must have no cycles.
- The Stable Dependencies Principle (SDP): Depend in the direction of stability.
- The Stable Abstractions Principle (SAP): Abstractness increases with stability.

**Acyclic Dependencies Principle - ADP**

Historical roots on typical issues of team-based development: the *morning after syndrome*[1] and the *weekly build*.

Solution to the problem: partition the development environment into releasable components
Each component has its own responsible developer/team so the releases are uniquely identified and components follow their own, private, development path. E.g. in git versioning system, using separated branches or even separated repositories.




- **Advantages**: Changes made to one component do not affect other teams, until they decide to adapt to the new component's release. I ntegration happens in small steps
- **Implications**: teams must monitor, manage and, maintain the dependency structure over time. there can be no "cycles" or circular dependencies in the dependency graph.

**Example: directed acyclic graph** Starting at any component, it is not possible to get back to the origin by following the edges (dependencies).

What happens when a change is made to Presenters? View and Main will be affected and tests with Interactor and Entities.

The bottom-up "building" for system release

- Entities
- Database and Interactor
- Presenters, View, Controllers, Authorizer
- Main

**Example 2: dependency cycle** Now imagine a new dependency arises: User class in Entities uses the Permissions class in Authorizer.

Testing Entities requires now also Authorizer and Interactor.

There are different ways to solve this problem:

- Solution A: Use dependency inversion principle (DIP)
- Solution B: Create a new component on which both Entities and Authorizer depend. Move the class(es) they both depend on into this new component.

[1]The morning after syndrome occurs when code worked on, tested, and left working one day is broken the next morning due to changes made by others

### 5.2.3 The Stable Dependencies Principle - SDP

Depend in the direction of stability. Stability is not about the frequency of changes, but about the amount of work required to make a change.

A cupboard is stable because it takes a lot of effort to move it, while a cup is not stable because it takes little effort to move it.

A component with lots of incoming dependencies is very stable because it requires a great deal of work to reconcile any changes with all the dependent components.

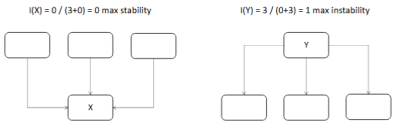
Modules that are designed to be harder to modify should not depend on modules that require little effort to change.

#### A simple measure of (in)stability

X is very stable because three components depend on it and it depends on no other component, while Y is very unstable because no component depends on it and it depends on three other components.

**Instability:** goes from 0 to 1, where 0 is the most stable and 1 is the most unstable.

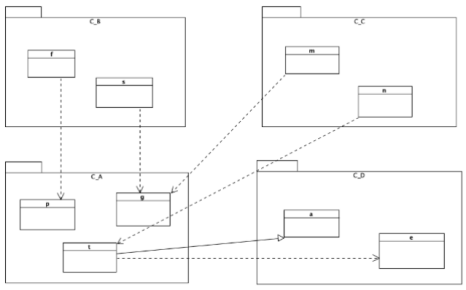
$$I = \frac{\text{Fan-out}}{\text{Fan-in} + \text{Fan-out}}$$




Where

- Fan-in is the number of incoming dependencies
- Fan-out is the number of outgoing dependencies

**Example:** I(C_B) = 0.33. Fan-in = 4 and Fan-out = 2. I(C_D) = $\frac{2}{4+2} = 0.33$.

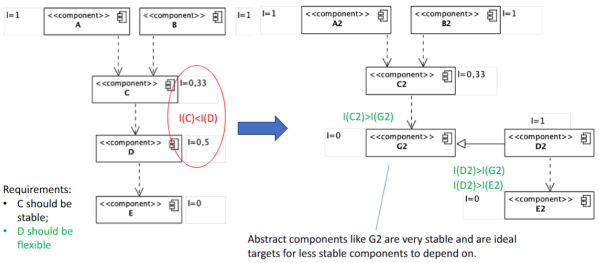


#### SDP and stability measure

Depend in the direction of stability.

The I(component) should be larger than the I(the components that it depends on). I metric should decrease in the direction of dependency. $I(C_B) = I(C_C) > I(C_A) > I(C_D)$

**SDP violation as system evolves: example** Requirements: C should be stable. D should be flexible.




### 5.2.4 Stable Abstractions Principle - SAP

A component should be as abstract as it is stable. Code that represent high-level policies of the system should be placed into stable components.

**SAP, OCP, SDP and abstract classes**

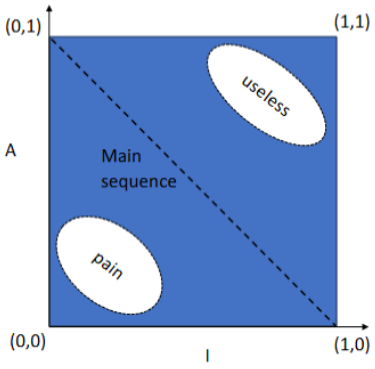
Open Closed principle suggests creating classes that are flexible enough to be extended without requiring modification. Abstract classes are best suited to this need. Thus, applying SDP: stability goes into the direction of abstraction

**A simple measure of abstractness this ratio:**

$$A = \frac{\text{Number of abstract classes and interfaces in the component}}{\text{Total number of classes in the component}}$$

where:

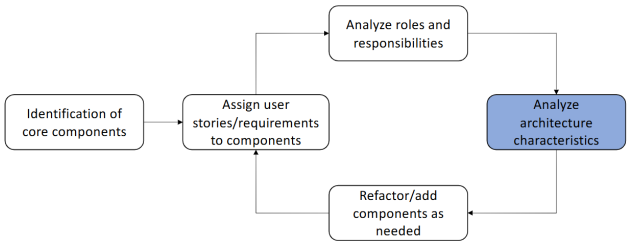
- A = 0 means that the component is completely concrete. No abstract classes.
- A = 1 means that the component is completely abstract. Only abstract classes.

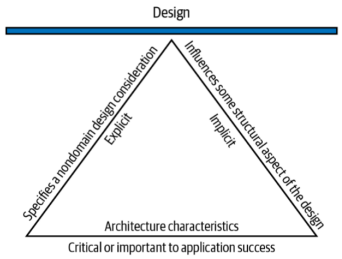



**Note:** Distance from the main sequence: $D = |A + I - 1|$

## 5.3 Strategic decisions

### 5.3.1 Iterative Refinement




**Operational Architectural Characteristics: examples**

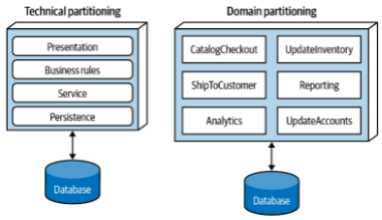
- **Scalability**: the system's ability to perform and operate as the number of users or requests increases
- **Elasticity**: the system's ability to perform and operate with peaks of requests
- **Responsiveness**: the system's ability to complete assigned tasks within a given time
- **Reliability/safety**: Whether the system needs to be fail-safe, or if it is mission critical in a way that affects lives. Fault tolerance does the software operate as intended despite hardware or software faults.
- **Availability**: How much of the time the system will need to be available; if that's 24/7, steps need to be in place to allow the system to be up and running quickly in case of any failure.
- **Continuity**: The system's disaster recovery capability.
- **Robustness**: The system's ability to handle error and boundary conditions while running, for example, if the internet connection or power fails.
- ...

**Engineering Architectural Characteristics**

- **Maintainability**: the degree of effectiveness and efficiency to which developers can modify the software to improve it, correct it, or adapt it to changes in environment and/or requirements.
- **Testablity**: how easily developers and others can test the software? How complete are tests?
- **Deployability**: ease, frequency and overall risk of deployment
- **Evolvability**: system's ability to survive changes in its environment, requirements and implementation technologies
- ...

### 5.3.2 Strategic level: partitioning

Technical partitioning vs domain partitioning




### 5.3.3 Conway's law

Organizations which design systems...are constrained to produce designs which are copies of the communication structures of these organizations.

The structure of the design will reflect how people communicate and work together

### 5.3.4 Monolithic versus Distributed architecture

**Monolithic**: single deployment unit of all code

**Distributed**: multiple deployment units connected through remote access protocols

| **Monolithic** | **Distributed** |
|---|---|
| Layered architecture | Service-based |
| Pipeline architecture | Event-driven |
| Modular monolith | Space-based |
| Microkernel | Service-oriented |
| | Microservices |

# Chapter 6

# Architecture visualization: C4 notation

## 6.1 Visualizing software architecture

There are not only one diagrams to visualize software architecture, but many of them.
Variety of goals, stakeholders, focus, etc.

## 6.2 Issues in sw architecture visualization

- Purpose of elements (boxes, arrows, etc.)
- Relationship between elements
- Levels of abstraction
- No context or logical starting point
- Consistency between diagrams
- Require explanations or descriptions to be fully understood

Diagrams are like maps: different providers, google or openstreetmap have different purposes. For this reason, we will introduce the C4 model, which is a standard for software architecture visualization.

## 6.3 C4 Model

With this model we can zoom-in and zoom-out to navigate through the different levels of abstraction to use with different stakeholders.

A **software system** is made up of one or more **containers**, each of which contains one or more **components**, which in turn are implemented by one or more code elements. And **people** use the software systems that we build.

It is organized in 4 levels of abstraction:

- Software System
- Container
- Component
- Code Element

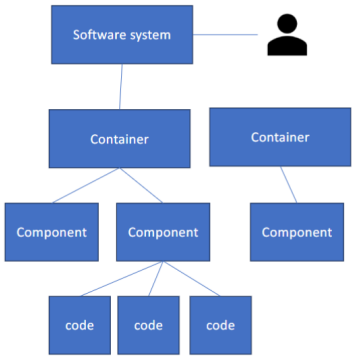



Figure 6.1: Static structure

### 6.3.1 Container

A boundary inside which some code is executed or some data is stored. Containers are **separately deployable**.

**Examples:**

- Application (server-side or client side)
- Database, file system, etc.
- Microservice (if not a system per se)

- Mobile app
- Script (python, shell, R, etc.)

### 6.3.2 Component

A grouping of related functionality encapsulated behind a well-defined interface
Components are **NOT** separately deployable and how components are packaged is not relevant at this level.

### 6.3.3 Code

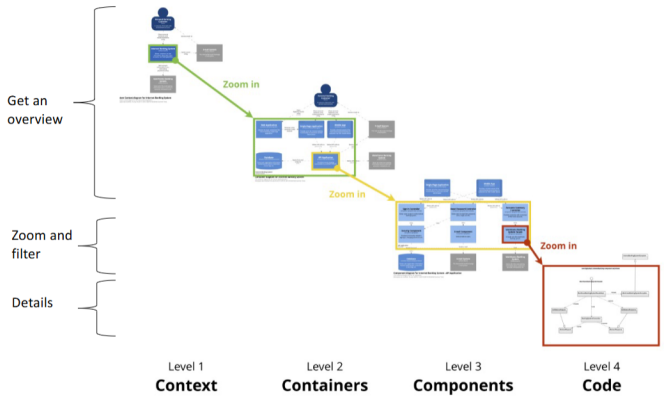
The basic building blocks of a programming language:

- classes,
- interfaces,
- functions,
- objects,
- enums

## 6.4 Different type of diagrams

The C4 model defines different types of diagrams for each level of abstraction.

- **Context diagram**: it shows how the software system in scope fits into the context around it.
- **Container diagram**: it shows the high-level technical building blocks (containers) and how they interact.
- **Component diagram**: it shows the components inside each container
- **Code diagram**: it shows how a component is implemented




Using the SPG, Solidarity Purchasing Group, presented before, as an example, we can create different diagrams for each level of abstraction.

## 6.5 Level 1: Context diagram

It helps answering such questions:

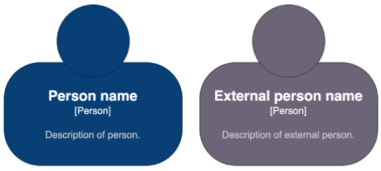
- What is the software system that we are building (or have built)?
- Who is using it?
- How does it fit in with the existing environment?

It is **required**: it is the starting point for communicating with any stakeholder
The key elements of a context diagram are:

- People who will use the software system
- Other software systems with which the reference system interacts
- + optionally the organization boundary

Interaction between elements: Directed arrows with high-level information about the purpose of that interaction. We can connect elements with arrows. It is better to use only one way arrows and not to use bidirectional once.

### 6.5.1 Information reported - People



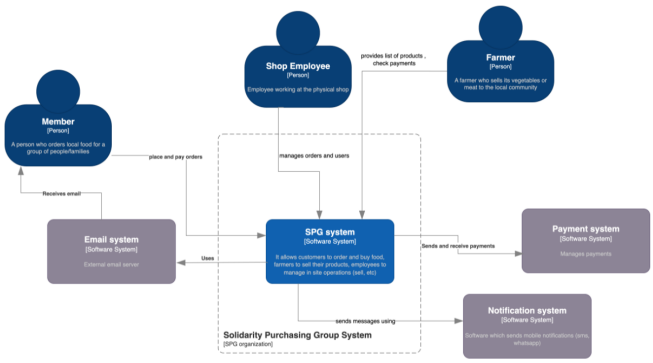

The people element have:

- **Name**: the name of the person, user, role, actor or persona.
- **Description**: A short description of the person, their role, responsibilities, etc

### 6.5.2 Information reported - Software system


The software system element have:

- **Name**: The name of the software system.
- **Description**: A short description of the software system, its goal and responsibilities, etc.

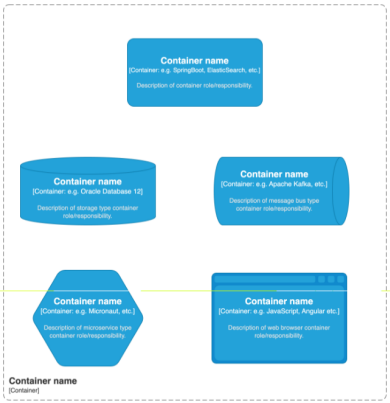



## 6.6 Level 2: Container diagram

Goal: illustrate the high-level technology elements and how responsibilities are distributed across elements and how containers communicate with each other

Every container must be deployable independently and the diagram is **required** and readed by software development adn operations stuff.

### 6.6.1 Elements




People, software systems and containers. For each container:

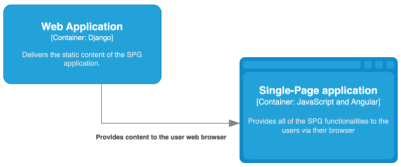
- Name: The name of the container (e.g. “Internet-facing web server”, “Database”, etc).
- Technology: The implementation technology (e.g. Java/Spring MVC application, ASP.NET web application, Relational Database Schema, etc.) or the general type of technology”
- Description: A short descriptive statement, or perhaps a list of the container’s key responsibilities or entities/tables/files/etc. that are being stored.

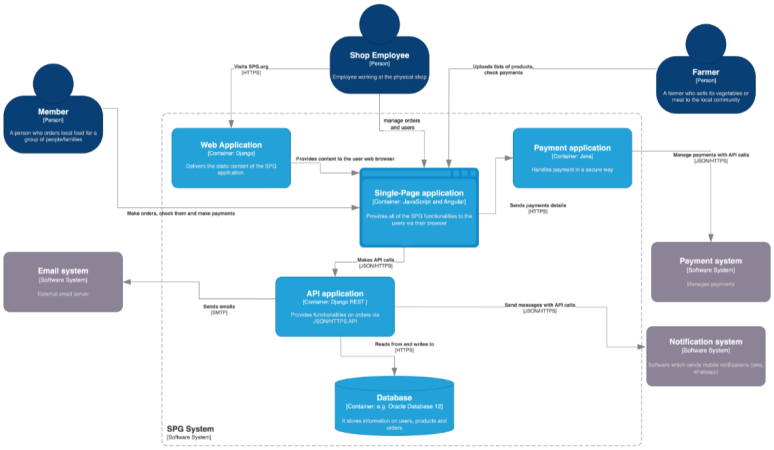
System boundary must be coherent with L1

### 6.6.2 Interactions between containers

Interactions between containers must be as explicit as possible.
Example of useful information to annotate the interactions with:

- The purpose of the interaction (e.g. “reads/writes data from”, “sends reports to”, etc).
- The communication mechanism (e.g. REST, Web API, Java Message Service, Web Services ).
- The communication style (e.g. synchronous, asynchronous, batched, twophase commit, etc).
- Protocols and port numbers (e.g. HTTP, HTTPS, SOAP/HTTP, SMTP, FTP, RMI/IIOP, etc)




## 6.7 Level 3: Components diagram

It shows components that are within a single container at a time. Infrastructural elements (e.g. , logging, metering) and shared library might be annotated, separated, or excluded.

It helps to answer the questions like

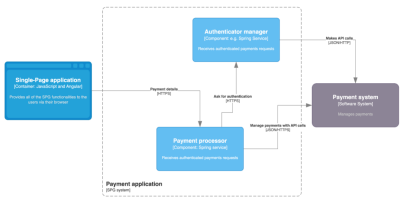
- What components is each container made up of?
- Do all components stay within in a container?
- How components interact each other ?
- How the software works at a high-level?

Optional, especially for containers like data storage, microservices. It is for technical people within the software development team.

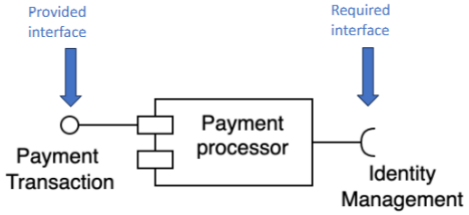
### 6.7.1 Elements

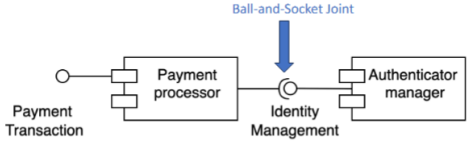
Same of the containers + components: people, software systems, containers and components.
Information to show for each component:

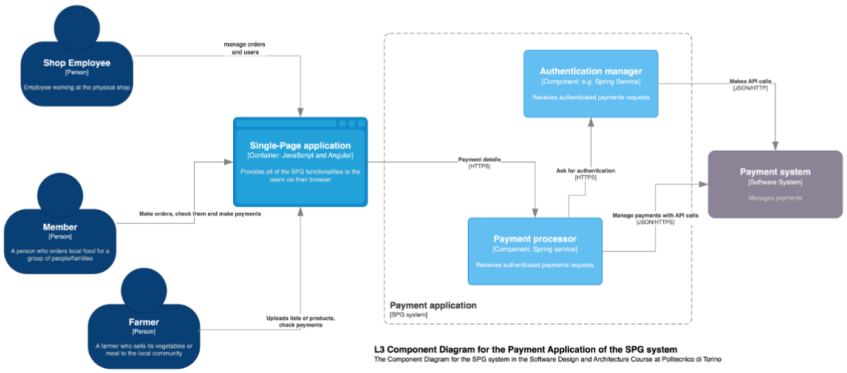
- Name: The name of the component.
- Technology: The implementation technology for the component (e.g. Java, Object, C#, Ruby).
- Description: A short description of the component in terms of its responsibilities




The UML component diagram is useful to provide more detailed documentation.

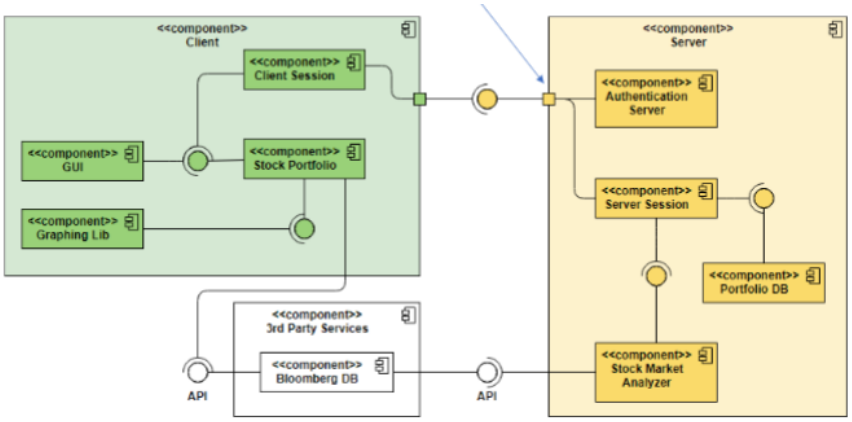




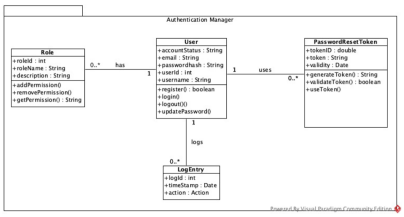

**Ports**

Ports indicate that required/provided interfaces are delegated to an internal class.




## 6.8 Level 4: Code diagram

It shows the implementation details of an individual component, it helps bridging the gap between architecture and code.
Usually, UML class diagrams are used beacuse:




- They can be generated automatically (e.g., with PlantUML)
- Large amount of details !

This is not mandatory, it is optional and it is especially for software developers, and more in general for technical people within the software development team.

> We can use Draw.io but no check on syntax. Instead we can use `https://academic.signavio.com/p/login`.
> Coerence between levels are important for exam.

## 6.9 Summary of C4 model

A lightweight notation to visualize the static structure of a software system It is based on a hierarchic conceptual model of software systems. 4 levels (2 of which mandatory).

> Four main elements, associated with unidirectional lines, directions shown is the most relevant.
> Customizable with colors , shapes, lines

- line styles for synchronous/asynchronous communication;
- colors for different components;
- different shapes for specific types of components;

# Chapter 7

# Wring Components

## 7.1 Components vs. Service

- **Component**: reusable piece of software that is used locally, eg static/dynamic linking. Is out of control of the writer of the component.

- **Service**: reusable piece of software that is typically accessed remotely, either synchronously or asynchronously, eg web service, RPC or soket.

## 7.2 Framework

A framework is a library that provides basic reusable assets that can be extended by the developers. Instead of writing from scatch, use some code that already exists.

It provides an abstraction layer over lower-level code, allowing developers to focus on higher-level features.

> **Note:** Extension and customization are often based on the inversion of control.

### 7.2.1 Inversion of control

"Don't call us, we'll call you"

With inversion of control, you code does not control the overall flow of the program. Instead, a framework or container controls execution and calls your code at the appropriate time.

#### Example of direct control

```
void main() throws IOException{
  Path path = Path.of("data/grades.csv");
  List<String> lines = Files.readAllLines(path);
  List<String> h = Arrays.asList(lines.get(0).split(","));
  processHeaders(h);
  lines.stream().skip(1)
  .map(r -> Arrays.asList(r.split(",")))
  .forEach(r -> processRow(r));

  double average = sum / (lines.size()-1);
  IO.println(average);
}
```

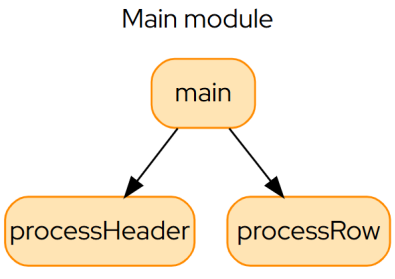



Figure 7.1: Call diagram

#### Example of inverted control

```
void main() throws IOException{
  CsvReader reader = new CsvReader();
  reader. headerProc(ReadCsvIoC: : processHeaders);

  reader. rowProc(ReadCsvIoC: :processRow);

  int n = reader.read("data/grades.csv");
  IO.println(sum/n);
}
```

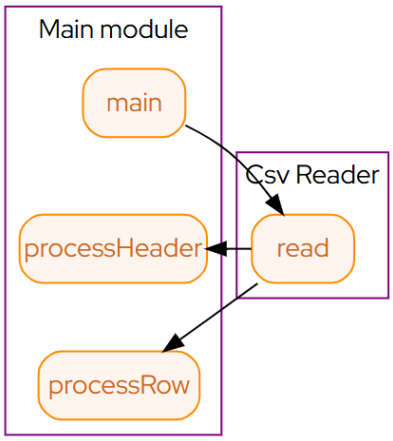



Figure 7.2: Call diagram

### 7.2.2 Patterns that enable inversion of control

The **Template method**, the **Strategy pattern**, and the **Observer pattern** are design patterns that enable inversion of control.

## 7.3 Wiring components

### 7.3.1 Interacction mode

- **Call-based**: function or method calls, the base logic of lots programming languages.
- **Event-based**: messages are sent and receaved. Used in GUI and web applications.
- **state-based**: state is updated and read from a shared memory. Used in embedded systems, writing in a specific memory address, and some peripheral devices read it.

### 7.3.2 Call binding

- **Source inclusion**: The code of the component is included directly in the main program during compilation. Strong implementation and compile-time dependecies.
- **Static linking**: Strong implementation dependency, limited compile time.
- **Dynamic linking**: Limited implementation dependency, limited compile time
- **Dynamic loading**: Strong construction dependency, no compile time

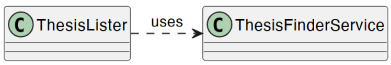
### 7.3.3 Method/funciton calls

Problem of simple calls:

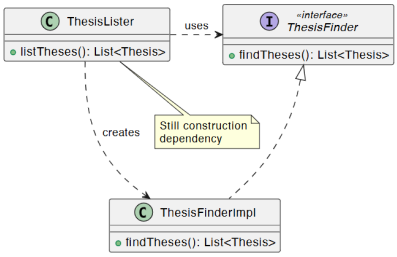
- Structural dependencies
- Runtime dependencies: spacial, temporal and synchronization dependencies
- Cascaing failures
- Limited scalability
- Hard to evolve independently

How put toghere components while minimizing the dependencies? E.g. they are build by different teams with little knowledge of each other.

**Example** A *ThesisLister*, frontend callses, provides a list of thesis matching a keykords using *ThesisFinder*.




Is not testable, not replaceble, heavy dependencies.
Using **interface** instead of concrete implementation to decouple the components, but still have dependencies.




## 7.4 Decoupling solutions

To solve the problem of dependencies, we can use 3 techniques for decoupling components :

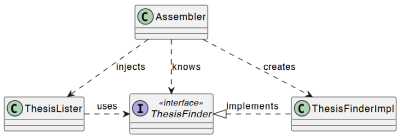
Dependency injection, Service locator, Message

### 7.4.1 Dependency injection

The *ThesisLister* calasss id dependent on both the *ThesisFinder* interace and the implementation of *ThesisFinderImpl*.

The goal of dependency injection is make the lister class unaware of the implementation class, but can still talk to an instance to do its work.

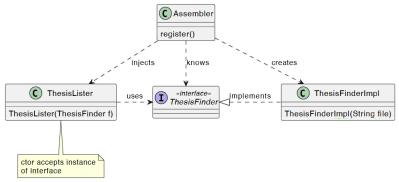
Move the logic of creating into an *Assembler* class




How to perfome the injection? Two tecniques:

- **Constructor injection**: use the constructor of the dependent class to inject the dependencies.
- **Setter injection**: objects are initialized with dependency using a setter

### 7.4.2 Constructor injection




#### Basic injection

```
ThesisFinder finder = new ThesisFinderImpl(``theses.csv''); //creates the implementation with parameters
ThesisLister lister = new ThesisLister(finder); //injects the dependency upon construction
```

#### Constructor injection with container

```
Container c = new Container();
Object[] parameters = {"theses.csv"}; //parameters for the implementation ctor
c.registerImplementation(ThesisFinder.class, ThesisFinderImpl.class, parameters); // binding interface to
↪  implementation
c.registerComponent(ThesisLister.class); // component registration
```

Reflection is used to find out how to link.

#### Constructor injection with annotations

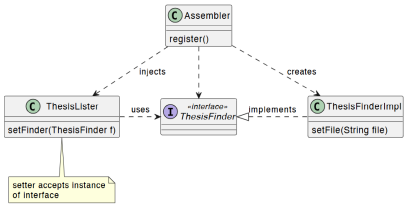
```
@Component
public class ThesisFinderImpl implements ThesisFinder {
  public ThesisFinderImpl(@Value("${thesis.file.path}") String file) {
    this. file = file;
  }
}

@Component
```

```
public class ThesisLister {
  public ThesisLister(ThesisFinder thesisFinder) {
    this. thesisFinder = thesisFinder;
  }
//...
}
```

> **Note:** The Spring framework is based on constructor injection with annotations.

### 7.4.3 Setter injection

Instead of using the constructor we can use a setter method to inject the dependencies.




#### Basic injection

```
ThesisFinderImpl finder = new ThesisFinderImpl(); //creates the server object
finder. setFile("theses.csv");  //set the parameter
ThesisLister lister = new ThesisLister()  //creates the client
lister.setFinder(finder); //injects the dependency with the setter
```

To set up a setting injection we can use a confriguation file.

#### Setter injection with configuration file

```
<beans>
  <bean id="ThesisLister" class="it.polito.
  ↪  ThesisLister">
    <property name="finder">
      <ref local="ThesisFinder"/>
    </property>
  </bean>
  <bean id="ThesisFinder" class="it.polito.
  ↪  ThesisFinder">
    <property name="file">
      <value>theses.csv</value>
    </property>
  </bean>
</beans>
```

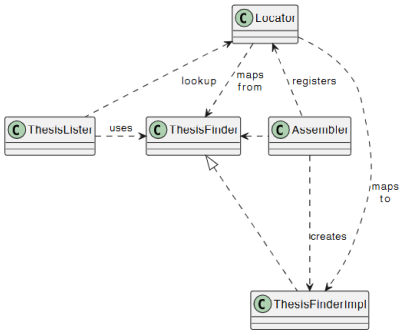
```
[
  {
    "id":"ThesisLister",
    "class":"it.polito. ThesisLister",
    "properties": [
      {"name": "finder", "local":"ThesisFinder"}
    ]
  },
  {
    "id":"ThesisFinder",
    "class": "it.polito. ThesisFinderImpl",
    "properties": [
      {"name": "file", "value":"theses.csv"}
    ]
  }
]
```

#### Setter injection with annotations

```
@Component
public class ThesisLister {

  private ThesisFinder finder;

  public ThesisLister() {}

  @Autowired
  public void setThesisFinder(ThesisFinder thesisFinder) {
  this. thesisFinder = thesisFinder;
  }
}
```

| Criterion | Constructor | Setter |
|---|---|---|
| Mandatory | Yes (object creation) | No (later stage) |
| Validity | Yes, always | Only after injection |
| Immutability | Yes (`final`) | No |
| Null safety | Fails fast | Risk later `NullPointerException` |
| Understandability | Depends explicit ctor | Depends hidden in setters |
| Circular dependencies | No | Can sometimes resolve |
| Optional dependencies | No, requires `Optional<>` or ctor overload | Yes (with `@Autowired(required = false)`) |

## 7.5 Service locator




A service locator is an object that knows how to get hold of all of the services that an application might need.
It is an alternative to dependency injection to break the dependencies.

**Example of locator service**

```
public class ThesisLister {
  public static String FINDER_SVC="thesis-finder";
  private final ThesisFinder thesisFinder;

  public ThesisLister() {
    this. thesisFinder = Locator.getLocator().lookup(FINDER_SERVICE, ThesisFinder.class);
  }
//...
}
```

**Basic assembler**

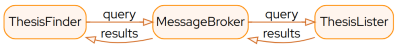
```
Locator l = Locator.getLocator();
l.register(ThesisLister.FINDER_SVC, new ThesisFinderImpl("theses.csv"));
ThesisLister thesisLister = new ThesisLister();
```

### 7.5.1 Locator vs. Dependency injection

| Criterion | Service Locator | Dependency Injection |
|---|---|---|
| Decoupling | Yes | Yes |
| Mechanism | Client explicit lookup | Provided externally |
| Inversion of Control | No | Yes |
| Visibility of dependencies | Hidden inside method calls to locator | Explicit in constructor/setters |
| Dependency on infrastructure | Client depends on the locator | No dependency on injector/container API |
| Ease of debugging | Easier (explicit calls) | Harder (framework wiring) |
| Simplicity | Straightforward | Less obvious at first |
| Reusability | Limited | Yes, no assumptions |
| Deferred linking | Yes, call locator anytime | No, injected at startup |

## 7.6 Messages/Events

Replace Direct call with a Broker




In this case sender does NOT know:

- Who consumes the message
- When the message is processed
- How many consumers exist

### 7.6.1 Roles: Producer, Consumer

| Producer | Consumer |
|---|---|
| Sends messages | Receives messages |
| Does not wait for processing | Processes independently |
| Does not know consumers | Can scale horizontally |

### 7.6.2 Roles: Broker

The broker, like in real life, is an intermediary that facilitates the communication between producers and consumers. It's a middleware responsible for:

- Storing messages
- Delivering messages to consumers
- Handling routing and filtering of messages

Some examples of brokers areRabbitMQ, ActiveMQ, Kafka, Amazon SQS.

### 7.6.3 Delivery Guarantees

It is possible to configure different levels of delivery guarantees depending on the requirements of the application.

| Type | Meaning |
|---|---|
| At most once | messages may be lost |
| At least once | messages may be duplicated, need idempotency |
| Exactly once | delivery guarantees, expensive |

Distributed systems always involve trade-offs.

#### Producer Example

```
ConnectionFactory factory = new ConnectionFactory();
factory.setHost("localhost");

try (Connection connection = factory. newConnection();
Channel channel = connection. createChannel() ) {

  channel.queueDeclare("orders", false, false, false, null);

  String message = "New Order: 123";

  channel.basicPublish("", "orders", null, message.getBytes());
  System.out.println("Sent: " + message);
}
```

#### Producer Example

```
ConnectionFactory factory = new ConnectionFactory();
factory.setHost("localhost");

Connection connection = factory.newConnection();
Channel channel = connection. createChannel();

channel.queueDeclare("orders", false, false, false, null);
```

```
DeliverCallback callback = (tag, delivery) -> {
  String message = new String(delivery.getBody());
  System.out.println("Received: " + message);
};

channel.basicConsume("orders", true, callback, tag -> {});
```

### 7.6.4 Structure

Producer and consumer are independent processes, they can run on different machines and the broker is infrastructure.

### 7.6.5 Shared state example

| Queue | Topic |
|---|---|
| 1 message → 1 consumer | 1 message → many consumers |
| Work distribution | Event broadcast |
| Load balancing | Publish/subscribe |

### 7.6.6 Effects

- **Construction decoupling**: producer does not create consumer
- **Spatial decoupling**: producer does not know location of consumer
- **Temporal decoupling**: consumer does not have to be active with producer. Producer does not have to block waiting for an answer
- **Data dependency**: data format and schema must be shared

### 7.6.7 Challenges

- Difficult debugging
- Eventual consistency
- Duplicate handling (idempotency!)
- Ordering issues
- Schema evolution

## 7.7 Shared state

Is a database just storage, or is it a communication mechanism?

Communication via mutations of shared, persistent state. The components communicate by reading and writing a common data store, no explicit message passing.

#### Shared state examples

- In-memory global variables
- Shared files
- Shared database
- Distributed cache (Redis)

#### Communication patterns

- State-based signaling
- Polling
- Work queue
- Derived state sharing

### 7.7.1 Effects

- No structural dependencies
- Low temporal dependencies: DB is like message broker
- Very high data dependency
- Simple mental model
- Easy querying and debugging

| **Aspect** | **Shared DB** | **In-memory Cache (Redis)** |
|---|---|---|
| Nature | Persistent system of record | Ephemeral shared memory |
| Latency | Higher | Very low |
| Consistency | Strong | Config-dependent |
| Visibility | High (queryable history) | Low (state disappears) |

# Chapter 8

# Communication

The communication between components can be implemented in different ways:

- Local vs Distributed
- Synchronous vs Asynchronous
- One-to-one vs One-to-many
- One-way vs Two-way
- Call vs Message vs Shared memory

There are some distributed computing fallacies that are worth mentioning:

1. The network is reliable;
2. Latency is zero;
3. Bandwidth is infinite;
4. The network is secure;
5. Topology doesn't change;
6. There is one administrator;
7. Transport cost is zero;
8. The network is homogeneous.

**Network latency** - Due to decomposition an operation may result into a series of round-trips between two services. Some solution can be: Batch API that allow feeding multiple requests or merge services to convert messages to method calls,

| | **Multiplicity** | |
|---|---|---|
| | **One-to-one** | **One-to-many** |
| **Sync** | Request/response | – |
| **Async One-way** | Notification | Publish/Subscribe |
| **Async Two-way** | Async. Req./Resp. | Publish/Async. Resp. |

### 8.0.1 One-to-one communication

**Request-response** - A client makes a request to a service and waits for a response. The client blocks while waiting.

**Asynchronous request/response** - A client sends a request to a service, which replies asynchronously. The client doesn't block while waiting.

**One-way notifications** - A client sends a request to a service, but no reply is expected or sent.

### 8.0.2 One-to-many communication

**Publish/subscribe** - A client publishes a notification message, which is consumed by zero or more interested services.

**Publish/async. response** - A client publishes a request message and then waits for a certain amount of time for responses from interested services.

## 8.1 Messaging

Messages are exchanged over message channels. A sender writes a message to a channel, and a receiver reads messages from a channel. The communication is asynchronous, the sender doesn't block waiting for a reply

**Header** The header contains metadata about the message, such as:

- Message ID
- Return address
- Metadata that describes the content of the message
- Collection of key-value pairs useful for routing, filtering, etc.

**Body** The body contains the data being sent in either text or binary format.

### 8.1.1 Message types

**Command** - This is equivalent of an RPC request. It specifies the operation to invoke and its parameters.

**Document** - Contains only data. The receiver decides how to interpret it, e.g. a reply to a command

**Event** - Indicates that something notable has occurred. An event is often a domain event, which represents a state change of a domain object.

## 8.2 Channel

Abstraction of the mechanism to transfer messages.

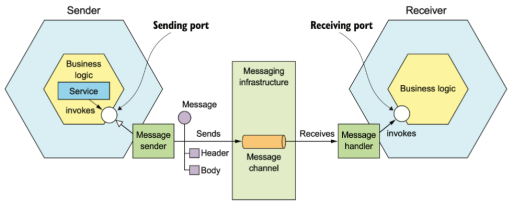
### 8.2.1 Point-to-point channels

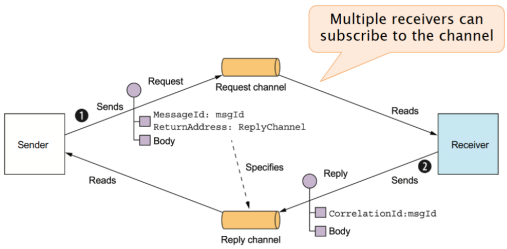
Delivers a message to exactly one of the consumers that is reading from the channel.

Services use point-to-point channels for the one- to-one interaction styles. E.g. command messages

### 8.2.2 Publish-subscribe channels

Channel delivers each message to all of the attached consumers. Services use publish-subscribe channels for the one-to-many interaction styles




## 8.3 Supported styles

- Request/response - supports synchronous or asynchronous
- One-way notification - No response
- Publish/subscribe - Multiple subscribers no response
- Publish/Async responses - Multiple subscribers, multiple responses

## 8.4 Direct messaging - brokerless

## 8.5 Broker-based messaging

Broker is an intermediary components through which all messages flow.

| Pros | Cons |
|---|---|
| Lighter traffic load, less latency | Services must be known *to* discovery |
| No centralized point of failure | Reduced availability, both must be available at the same time |
| Simplicity | Hard to have guaranteed delivery |

| Pros | Cons |
|---|---|
| Loose coupling. No need to know each other or to be active at the same time | Potential performance bottleneck |
| | Single point of failure |
| | Additional complexity |

### 8.5.1 Broker characteristics

It support both standard protocols, e.g. AMQP, MQTT, STOMP, or proprietary protocols. It has some good feature such as:

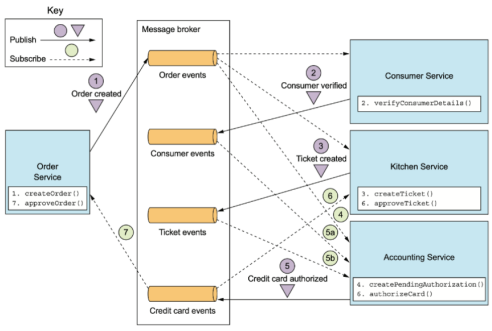
- Messaging ordering
- Guaranteed delivery
- Persistence
- Durability, consumer can disconnect and reconnect without losing messages
- Scalability
- End-to-end latency
- Support for cometing consumers

## 8.6 Coordination

### 8.6.1 Choreography

Each local transaction publishes domain events that trigger local transactions in other services.

- It requires reliable message infrastructure and messages with correlation id.
- Basic operation is simple, it perform operation and publish event
- Difficult to understand and maintain because transaction logic is scattered
- Complex service dependencies also cyclic, might lead to tight coupling

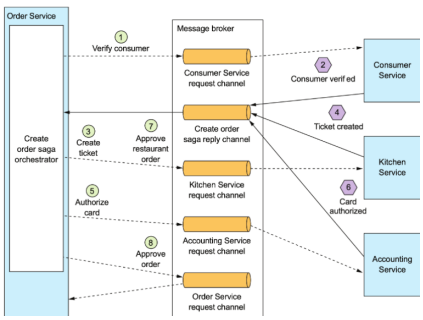


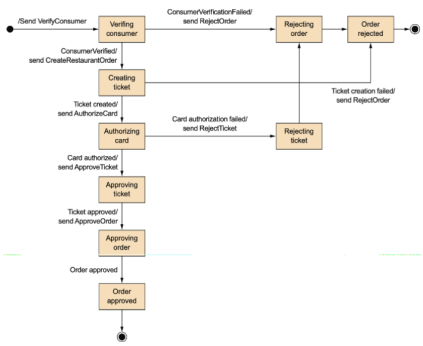

### 8.6.2 Orchestration

An orchestrator (object) tells the participants what local transactions to execute.

This teqnique requires reliable messaging, it has a simple dependency structure with reduced coupling.

The business logic is simple and services are not aware of the saga. Transaction logic is gathered in a single place, must be kept separate from service logic.

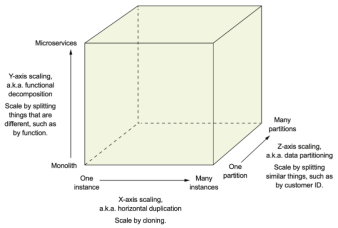


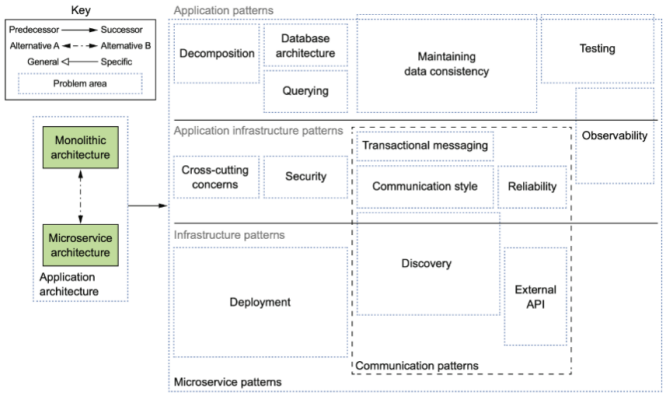
# Chapter 9

# Microservices

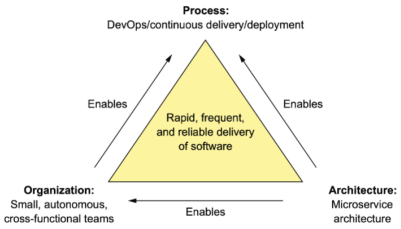
The scalability dimension of software architecture can be represented by the following figure:




| **Advantages** | **Disadvantages** |
|---|---|
| Enables the continuous delivery and deployment of large, complex applications | Finding the right set of services is challenging |
| Services are small and easily maintained | Distributed systems are complex, which makes development, testing, and deployment difficult |
| Services are independently deployable | Deploying features that span multiple services requires careful coordination |
| Services are independently scalable | Deciding when to adopt the microservice architecture is difficult |
| Enables teams to be autonomous | |
| Allows easy experimenting and adoption of new technologies | |
| Better fault isolation. | |



### 9.0.1 Monolith vs. Microservices



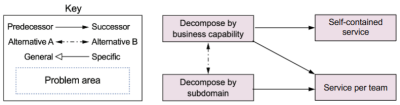

The problem is the deployment in fact:

- Monolithic architecture: Simple to develop, deploy, and scale.
- Microservices architecture: Simple to mantain, to test and flexible to scale

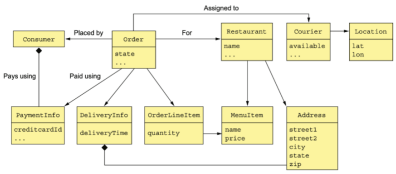
### 9.0.2 Decomposition

Decomposition identifies system operations:

- Understand domain model
- Define operations
- Identify use cases




and decompose the system into subsystems or subdomains.




**Domain model Example**

**User stories**

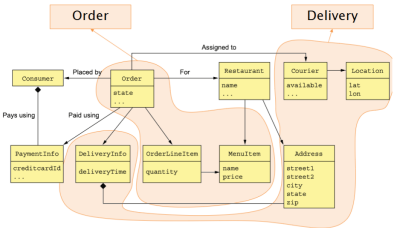
| Actor | Goal | Services |
|---|---|---|
| Consumer | Create order | Order |
| Restaurant | Accept order | Order |
| Restaurant | Complete order | Kitchen |
| Courier | Update location | Delivery |
| Courier | Pick up | Delivery |
| Courier | Delivery | Delivery |
| Delivery company | Manage couriers | Courier |
| Delivery company | Courier payment | Accounting |

**By business capability**

1. An organization business capability is what it does (as opposed to how). Look into organization structure, also consistent operations, stories or goals.
2. Single Responsibility Principle
3. Common Closure Principle - classes that change for the same reason should be in the same package
4. Result is cohesive and loosely coupled

**By subdomain**

- Identify bounded contexts - DDD sub-domains. Partitions of overall domain model
- Result is cohesive, Loosely coupled, allow cross-functional and autonomous teams and scalable architecture



### 9.0.3 Guidelines from OOP

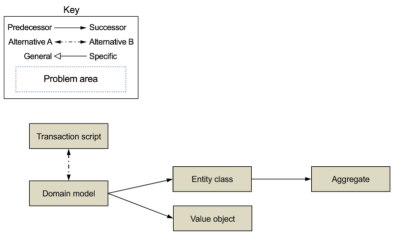
**Single Responsibility Principle** - A class should have only one reason to change.

**Common Closure Principle** - The classes in a package should be closed together against the same kinds of changes. A change that affects a package affects all the classes in that package.

**Decomposition obstacles**

- Network latency
- Reduced availability due to synchronous communication
- Data consistency across services
- Obtaining a consistent view of the data
- Good classes

## 9.1 Business logic




### 9.1.1 Transaction script

When you have simple business logic that consist of a sequence of steps.
Organize business logic by procedures where each procedure handles a single request from the presentation.
One service class with a method for each procedure. Several data classes.
Consequences:

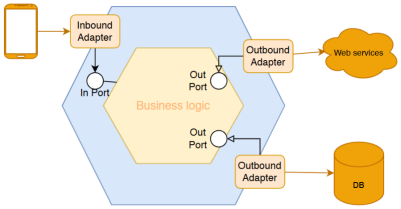
- All the code for a request in a single place
- Code duplication for similar requests
- Difficult to maintain and refactor

### 9.1.2 Domain model

When you have complex business logic
Adopt recommended object-oriented methods to create classes that encapsulate both code and data, using the DDD approach.
Consequences:

- Code for a request is spread among classes
- No code duplication
- Easy to maintain and refactor
- Easy to extend with Strategy or Template Method patterns

### 9.1.3 Ports & Adapters (hex)



## 9.2 Domain Driven Design - DDD

- Sub-domains
- Bounded context
- Class types patterns: Entity, Value Object, Service, Repository, Factory
- Aggregate

### 9.2.1 Sub-Domain & Bounded Context

DDD decompose the overall domain model into sub-domains. Each sub-domain is valid within a bounded-contex.

A bounded-context brings together a common language and people, team, focusing on a shared problem. Distinct contexts are related to each other in order to enable communication.

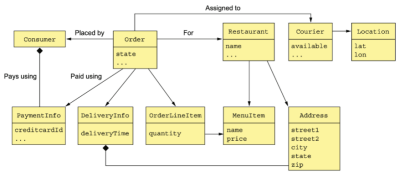
### 9.2.2 Class patterns

- Entity - Represents objects that have a persistent identity. Two objects with the same values represent distinct items. Usually, they are persisted to a db
- Value object - Represents a collection of values. Two objects with the same values are interchangeable
- Factory - Implement object creation logic. May hide concrete class
- Repository - Provides access to persistent entities. Hides access to db
- Service - Implements business logic that does not belong to a single entity

### 9.2.3 Domain boundaries

Often domain models have fuzzy boundaries. All strictly related concepts should belong to the same context. E.g. `Order` and `OrderLineItem` classes.

Not keeping together such classes might result in violations of business rules or invariants, upon updates. E.g. minimum amount for an order.

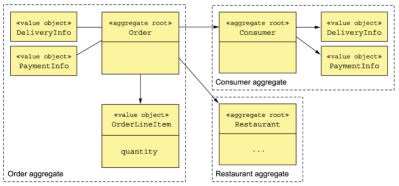



### 9.2.4 Aggregate pattern

Organize a domain model as a collection of aggregates, each of which is a graph of objects that can be treated as a unit.

Decompose domain model into chunks:

- Self contained units
- Easier to understand
- Clarify scope of operations
- All elements share the same life cycle



### 9.2.5 Rule: reference roots only

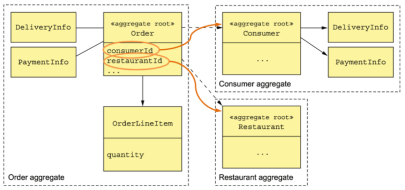
Reference only the aggregate root. Updates go through methods of aggregate root. When a repository loads an aggregate from the db it returns a reference to the aggregate root.

Consequences:

- Enable aggregate to enforce consistency rules
- Aggregate updates are serialized

### 9.2.6 Rule: primary keys inter aggregate

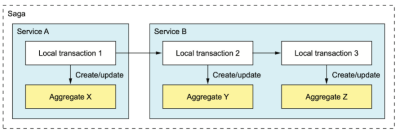
Inter-aggregate references **must use primary keys**. NOT using object reference, indeed, this is usually code smell.
Consequences:




- Loose coupling of aggregates
- Keep aggregate boundaries enforced
- Avoid accidental cross-aggregate updates
- Aggregates can belong to distinct services
- Simplify persistence

### 9.2.7 Rule: one aggregate per transaction

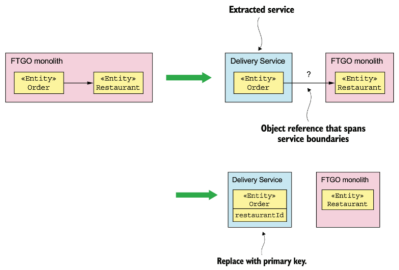
One transaction operates on one aggregate only
Consequences:



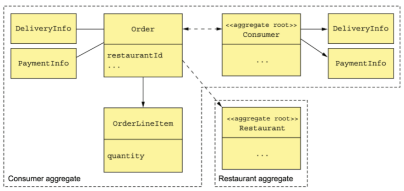

- Ensure a transaction is contained in a single service
- Suitable for limited transactions support in NoSQL databases
- Less efficient in transfer from/to db
- More complex implementation of complex requests that update multiple aggregates

## 9.3 Split monolith with aggregates

Transformation into aggregates can help splitting the monolith. Services can be extracted easily.




### 9.3.1 Aggregate granularity




- Fine grained: Improve scalability (Updates on an aggregate are serialized), reduce the risk of conflicts and it is easier to decompose into services
- Coarse grained: More suitable to fit all relevant entities in a transaction

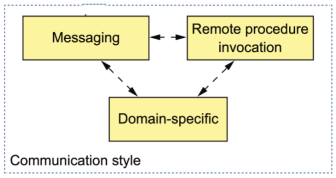
### 9.3.2 Domain Events

An Aggregate publishes a Domain Event when it's created or undergoes some other significant change. Represented by a class in the domain model, the name is formed using a past-participle verb phrase. E.g. `OrderPlaced`, `OrderShipped`, etc.

Properties that convey the event meaning. Each property is either a primitive value or a value object, also metadata, e.g., event ID, timestamp

- Maintaining data consistency across services
- Notifying a service that maintains a replica that the source data has changed
- Sending notifications to users
- Notifying another service to update related data
- Notifying a different application via a registered webhook or via a message broker
- Monitoring domain events to verify that the application is behaving correctly.
- Analyzing events to model user behavior

## 9.4 Communication




Ensure agreement on service interaction mechanism to avoid late integration issues:

- Define the interface (using an IDL)
- Review with client services' developers
- Iterate to reach consensus
- Start implementation of services

### 9.4.1 API evolution

"An interactive software system must be continually adapted or it becomes progressively less satisfactory".
Aim for minor backward-compatible changes.
Robustness principle:

- Be conservative in what you do
- Be liberal in what you accept from others

## 9.5 Message formats

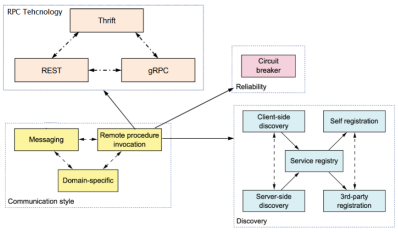
Text-based formats: JSON, XML, YAML

Binary formats: Protocol Buffers, Avro, Thrift

Once the format is chosen, it might be mandated by the communication mechanism

## 9.6 Remote procedure call (RPC)

- **Pros**: **Simple** and familiar, request-response is easy and the system is simple because of no intermediate broker
- **Cons**: Usually **only supports request/reply** and not other interaction patterns such as notifications, request/async response, publish/subscribe, publish/async response. **Reduced availability** since the client and the service must be available for the duration of the interaction




Requires service discovery

### 9.6.1 Robust Synchronous APIs

- Network timeouts: Never block indefinitely, always use timeouts when waiting for a response
- Limit outstanding requests: Define an upper bound on the number of outstanding requests that a client can make to a particular service
- Circuit breaker pattern, allow failing fast

### 9.6.2 Circuit Breaker pattern

Service should be invoked through a proxy. Track the number of successful and failed requests. If the error rate exceeds a threshold, trip the circuit breaker so that further attempts fail immediately. After a timeout period, the client should try again, and, if successful, close the circuit breaker.
This is implemented by Netflix's Hystrix library.

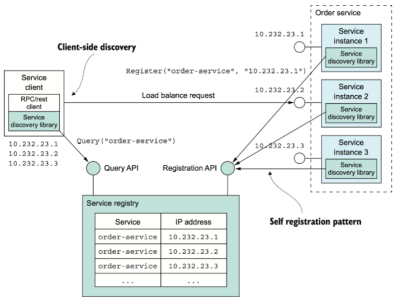
### 9.6.3 Service Discovery

Using RPC you need to know the network location of your service.
In microservices architectures service instances change continuously due to:

- Autoscaling
- Failures
- Updates

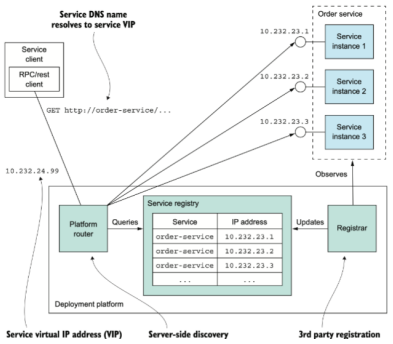
**Application level discovery**

- Self registration pattern: service register itself on activation and update list (Health check endpoint on server, Heartbeat API callback on registry)
- Client-side discovery pattern: Queries registry for active servers. Uses load balancing algorithm to pick one.




**Platform level discovery**

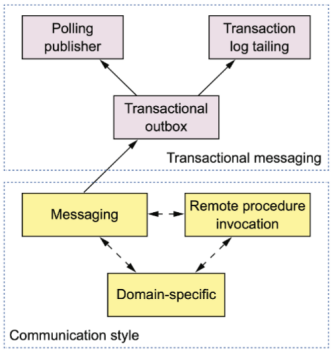
- Self registration pattern
- Third party registration pattern: Registrar component perform registration, often part of the platform (e.g., Kubernetes or Docker)
- Server-side discovery pattern: Client makes DNS request. Router access registry, load balance and resolve DNS



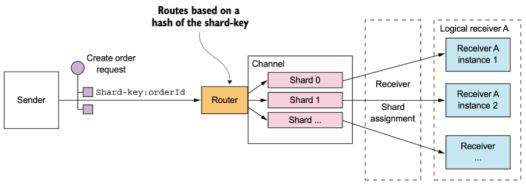

## 9.7 Messaging

### 9.7.1 Competing receivers

- Multiple receivers
- Each message must be processed only once or processed in order
- Sharded (partitioned) channels. The sender specifies a shard key in the message's header. The message broker uses a shard key to assign the message to a particular shard/partition.




## 9.8 Shared channels




### 9.8.1 Duplicate messages

Failures of client, server, or broker might represent a problem. Message are guaranteed to be delivered at least once, but not only once. Possible countermeasures:

- Idempotent messages
- Message tracking

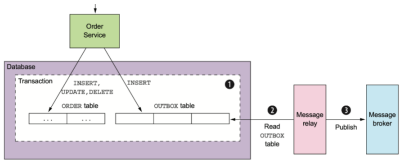
### 9.8.2 Transactional messaging

Use a db table as a message queue. Keep in the same transaction:

- Db application update
- Message queueing

#### Transactional outbox

The outbox table acts as a message queue within the database.




1. The transaction updates the order table and inserts a message into the outbox table.
2. A message relay reads the outbox table.
3. The relay publishes the message to the message broker.

## 9.9 Data and Transactions

### 9.9.1 Database per service

Services must be loosely coupled so that they can be developed, deployed and scaled independently. Each service's persistent data is private and accessible only via its API, options:


- Private-tables-per-service
- Schema-per-service
- Database-server-per-service

A service's transactions only involve its db.

### 9.9.2 Transactions in a monolith

ACID db transaction : Atomicity, Consistency, Isolation, Durability. Lack of isolation leads to:

- Lost updates
- Dirty reads
- Nonrepeatable reads

#### Example: confirm order

- **Order Service:** Create an Order in an APPROVAL_PENDING state.
- **Consumer Service:** Verify that the consumer can place an order.
- **Kitchen Service:** Validate order details and create a Ticket in the CREATE_PENDING state.
- **Accounting Service:** Authorize consumer's credit card.
- **Kitchen Service:** Change the state of the Ticket to AWAITING_ACCEPTANCE state.
- **Order Service:** Change the state of the Order to APPROVED state.

### 9.9.3 Distributed data consistency

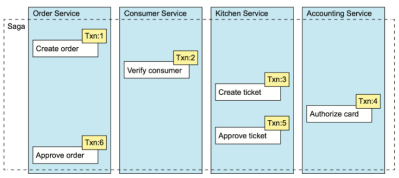
- **Distributed transaction - 2PC:** Query to commit, Agree/Abort, Commit/Rollback, Acknowledge.
- **Saga:** Local transactions coordinated with messages.

#### Distributed transactions limits

Transactions are not universally supported (e.g. NoSQL dbs, Modern message brokers). Transactions are synchronous, which means reduced availability. Isolation is overrated (most dbs adopt weaker isolation levels). Impossible to have both availability and most recent data.

### 9.9.4 Saga pattern

A saga is a sequence of local transactions. Each local transaction updates the database and publishes a message or event to trigger the next local transaction in the saga. If a local transaction fails then the saga executes a series of compensating transactions (undo the changes that were made by the preceding local transactions).



#### Compensating transactions

Rollback is not directly supported since all service transactions are separate. Explicit invocation of compensating transactions in case of rollback (Used, e.g., when a check in a step fails).

#### Saga transactions types

- **Compensatable transaction:** Can potentially be rolled back using a compensating transaction.
- **Pivot transaction:** The go/no-go point in a saga. If succeeds the saga can run until completion.
- **Retriable transaction:** Follow the pivot transaction and is guaranteed to succeed.

#### Saga non-isolation

- **Atomic:** Managed by coordination mechanism and compensating transactions.
- **Consistent:** each service guarantee consistency in its own transaction.
- **Durable:** ensured by local db.

Non isolated problems: Lost updates, Dirty reads, Non-repeatable reads.

#### Countermeasures for non-isolation

- **Semantic lock:** An application-level lock. A compensatable transaction sets a flag in any record that it creates or updates indicating it isn't fully committed. It's cleared by either a retriable transaction or a compensating transaction.
- **Commutative updates:** Design update operations to be executable in any order (e.g., `Account.debit()` decrement balance).
- **Pessimistic view:** Reorder the steps of a saga to minimize business risk due to a dirty read. Cancel order (rolled back) and create order when executed concurrently could lead to exceeding customer's credit.
- **Reread value:** Prevent dirty writes by rereading data to verify that it's unchanged before overwriting it. If changed, saga aborts.
- **Version file:** Record the updates to a record so that they can be reordered (turns noncommutative operations into commutative).
- **By value:** Use each request's business risk to dynamically select the concurrency mechanism (sagas vs distributed transactions).

### 9.9.5 Domain event pattern

An aggregate publishes a domain event when it's created or undergoes some other significant change. A domain event is a class with properties that meaningfully convey the event, a name formed using a past-participle verb (e.g. `OrderCreated`), and metadata (e.g. event ID, timestamp).

### 9.9.6 Event sourcing

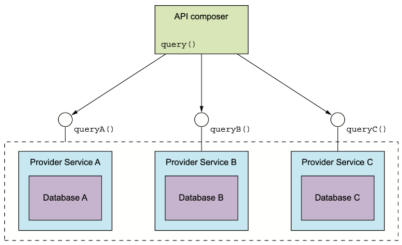
Event sourcing is an event-centric technique for implementing business logic and persisting aggregates. An aggregate is stored in the database as a series of events, where each event represents a state change. It is an alternative to common persistence, e.g. through ORM.

| Pros | Cons |
|---|---|
| Reliably publishes domain events | It has a different programming model that has a learning curve |
| Preserves the history of aggregates | Messaging-based application are complex |
| Avoids the O/R impedance mismatch | Evolving events can be tricky |
| Provides developers with a time machine | Deleting data is tricky |
| | Querying the event store is challenging |

## 9.10 Queries

Queries can be Simple (require data from a single $\mu$S db) or Composed (required data from multiple $\mu$S dbs).

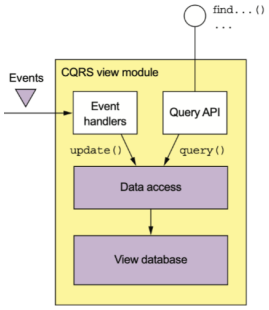
### 9.10.1 API Composition



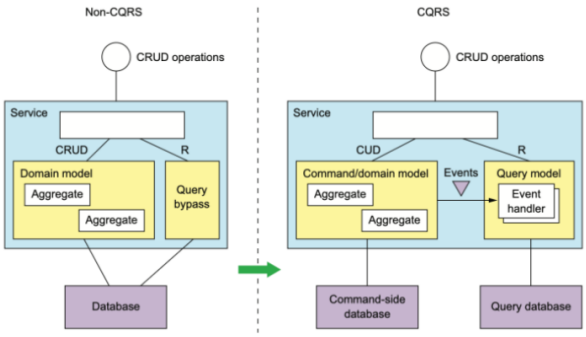

- **API Composer:** Implement a query that retrieves data from several services by querying each service via its API and combining the results.
- **Provider Service:** Implement a service-limited query, limited by how data is partitioned and capabilities of APIs/databases.

Composer role can be Client (e.g. web app backend), API Gateway, or a Standalone service. For efficiency, parallelize queries as much as possible.

### 9.10.2 CQRS



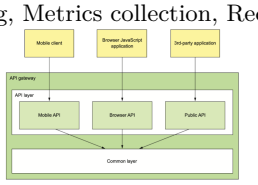

**C**ommand - **Q**uery **R**esponsibility **S**egregation. Use events to maintain a read-only view that replicates data from the services. The view can be easily queried.
Design the schema to match the queries. The data access module must handle idempotent updates and concurrent updates. Implement a mechanism to efficiently build or rebuild the view, and cope with the replication lag.



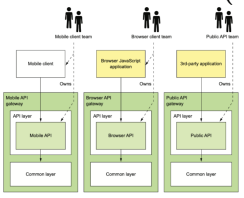

## 9.11 External API

External clients have low bandwidth (negative impact on UX) and lack of encapsulation. Different communication mechanisms are used. Possible solutions:

**Edge Functions** Authentication, Authorization, Rate limiting, Caching, Metrics collection, Request logging.



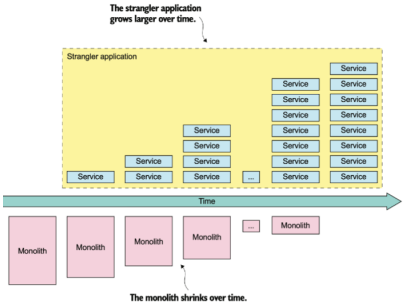
**Backend for Frontend (BFF)**



## 9.12 Transition to Microservices

- **Big-bang rewrite:** "the only thing a Big Bang rewrite guarantees is a Big Bang!" (M.Fowler) .
- **Strangling the monolith:** Progressively wrap the monolith until it becomes an empty shell.

### 9.12.1 Strangling the monolith




**Context**: a monolith application should be migrated to a micro service one
**Problem**: a complete big-bang rewrite poses several risks
**Solution:** gradually create a new system around the edges of the old, letting it grow slowly over several years until the old system is strangled.
**Consequences:** Reduced risks, frequent releases allow to monitor the process and provide steady release of value, speed of transition can be adjusted.

### 9.12.2 Transition strategies

- **Implement new features as services:** Stop the growth of the monolith. API gateways reroute requests. Integration glue code integrates the service with the monolith (typically adapters).
- **Separate presentation and backend:** Represents an intermediate strategy.
- **Break up the monolith:** Extract functionalities into services. Extract a vertical slice (Inbound adapters, Domain logic, Outbound adapters, Monolith's database schema).

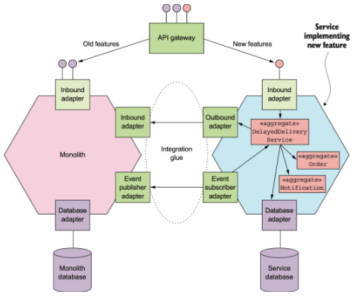



Figure 9.1: New features as services

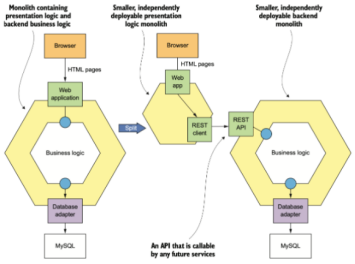



Figure 9.2: Split separation and backend

### 9.12.3 Refactor DB

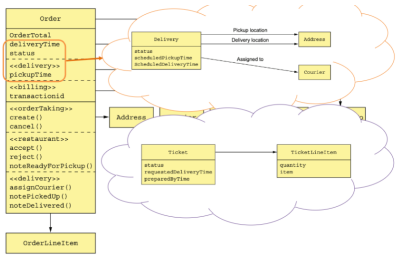
Several classes are persistent so the relative tables in the db must be refactored. Replicate data to avoid widespread changes in the monolith (Keep original schema and use triggers to update data).

**Selecting services to migrate** Rank modules by anticipated value your organization can get by migrating:

- Features providing new business capabilities
- Accelerated development
- Performance, scaling, reliability issues
- Unlock migration of other more valuable modules

### 9.12.4 God Class

An object that controls way too many other objects in the system and has grown beyond all logic to become The Class That Does Everything. Often implements business logic for many different aspects.
**Splitting the God Class:** Apply DDD to separate domain models into subdomains. Avoid tight coupling among services and avoid a data service. Different views must be kept consistent (via Saga) and UX must be adapted to the set of partial models (via API gateway).



# Chapter 10

# Bending Spoons seminary (Evernote)

Founded in 2013 in Danmark. They acquire digital products then unlock the potential of the products.

500+ TB data processed per year. Constantly search for new talent.

| They look for: | The processes: |
|---|---|
| Drive | Screening |
| Potential | Tasks |
| Passion | Interviews |
| | Final decision |

## 10.1 Evernote

There is a frontend that get the data from bakend at the very high level, but deeper the system is very complex and full of components.

Before the acquisition Evernote was a monolith, but after the acquisition they decided to migrate into a set of microservices with each one thake care of a specific element.

All of there service shoudl minimize the dependecies with the other services.

But the backend is not the complex part of the system.

### 10.1.1 Evernote's sync system

Legacy: "Monolith have you some new content?" "Yes, the note with the monkey" This is a polling request useless for the user and for the backend system.

Microservices: Persistent connection between the client and the server, faster then polling, cheaper for both: the client side save battery and for the server side save CPU and bandwidth.

#### Deep dive into sync activity

Note creation:

- The phone perform an http request to the backend to note service
- Performing some checks
- Create the json file SQL database
- Note create message to the sync service queue
- The sync service send the message to the all the client devices with the new note

#### Client connection

Fernando autenticate withe the websocket and then connect to sycn service. This service has tons of persistent connection always open, it use a multi-map, because of multiple devices but same user, to keep track of the connection and the user id.

The persistent connection can go offline for a bit. Do they loose all the message? Yes but the sync service use google cloud datastore. It is a single schema with the UId, the Instance with every metadata of Fernando and SyncOperation with the operation performed (CREATE, UPDATE, DELETE).

Every time the client receaves some message, it reaceve the userId and the timestamp and this is used in the google cloud datastore.

```
SELECT * FROM sync_documents
WHERE accessInfo.updated >= "1259414" and accessInfo.userId = "12345"
```

### 10.1.2 The sync service

| Pros | Cons |
|---|---|
| Fast updates | Data duplication |
| Offline access | Integration difficulty |
| Cheap | Complexity |

# Chapter 11

# Boundary

Architecture is defined by the boundaries drawn within a system and the dependencies that cross those boundaries, rather than the physical mechanisms through which components interact and execute functions.. Boundaries between components allow to make “firewalls” against harmful propagation of changes. The components on one side of the boundary change at different rates and for different reasons than the components on the other side of the boundary

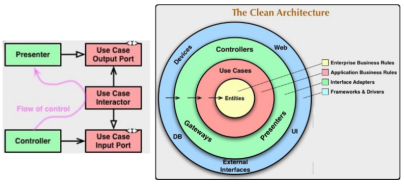
## 11.1 Drawing boundaries

1. Partition the system into components
2. Identify core business components from those containing functions not related to the core business
3. Arrange the code so that dependencies point toward the core business

Make use of SOLID principles and Component principles.

### 11.1.1 The clan architecture

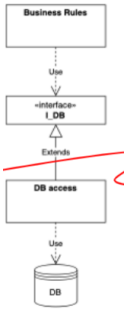
- Outer circles: mechanisms.
- Inner circles: policies.




Dependency Rule : source code dependencies point only inward, toward inner circles (i.e. higher-level policies)

### 11.1.2 Independence and Decoupling of layers

Layers allow independent development, deployment, operation, and maintenance. Two type of layers: Horizontal (DB, UI, etc.) and Vertical (Source code units, services, etc.)

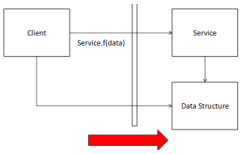



Decoupling of policy from the details. Defer decisions on details for as long as possible.
DB component knows about use case rules component, but NOT viceversa. **Low level components know** everything of the **higher-level** components, but NOT viceversa.
The Database component contains the code that translates the calls made by the BusinessRules into the query language of the database. This translation code knows about the BusinessRules. Business rules can be adapted to any kind of DB.

## 11.2 Crossing boundaries

Are required for: require a service, ask for data or foward requests. Only data cross boundaries, not entities. Data is passed in the most convenient format for the inner circle. Entity objects or database rows are NOT passed, otherwise Dependency Rules mught be violated.

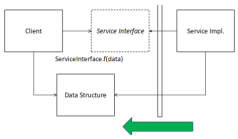
#### Low-level client to high-level server




The Client calls a function f() on the Service. Data is passed as argument of the function. Dependencies cross boundary from left to right point toward the higher-level component.

**High-level client to low-level server**

The Client calls the f() function of the ServiceImpl, through the Service interface. Dependencies cross the boundary always toward the higher- level component.




### 11.2.1 Data is significant. Data storage is a detail

From an architectural perspective, the database is NOT an entity. It is a detail that does not rise to the level of an architectural element.

It is an architectural mistake to allow database rows and tables to be passed around the system as objects. This is because it causes the use cases, business rules and, in some cases, also the UI to be coupled to the relational structure of the data.

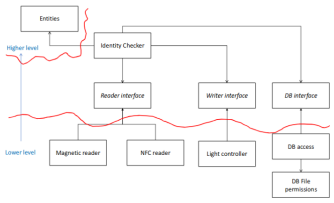
The the data model, i.e. the organizational structure of data, instead, is architecturally significant.

**Example** In our university, we need a software that manages the identities. One of the use cases is to check identity from the polito badge and grant/deny access for a specific area or room of the university. It should be possible to check whether a person (student, professor, etc) have access to a given laboratory in each department.

Policy about permissions (including which days and times, and for how long time) are defined on a specific database. Card readers use the magnetic stripe, but soon there will be also NFC card readers in some department accesses. A red or green light on the reader will show the result of the output

**Key aspects**

What is policy and what details ? Rreading bages, database, rules, output
What is low level and what is high level ?

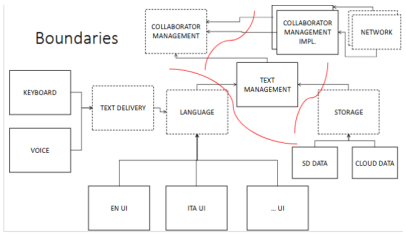



## 11.3 Layers

The most basic components of a system are:

- User Interface (UI)
- Application Logic
- Database

**Example: a word processor** A word processor (WP) is a device or software for creating, editing, storing, and printing text documents.




## 11.4 Test

Tests support development, not operation, HOWEVER They are an integral component of the system, akin any other elements within the system participate in its architecture. They are part of the design of the system architecture. Also tests adhere to the Dependency Rule.

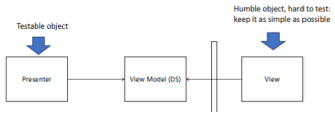
### 11.4.1 Design for testability

Tests are coupled to other system's elements. Changes to components in the inner circles may break several tests, to reduce tests fragility, tests should depend as little as possible to volatile elements.

Example: business rules should be tested without using the GUI. Specific tests API can avoid constraints (e.g. security).

### 11.4.2 Humble objects

The Humble Object helps unit testers to separate behaviors that are hard to test from behaviors that are easy to test. It is adopted in software architecture to identify and protect architectural boundaries.



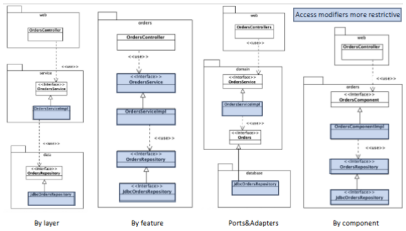

Esample are: Database gateways, Data mappers and Service listeners.

## 11.5 Architectural styles

Examples of SPG/order.
Use case: customers should be able to view the status of their orders
Classes:

- OrdersController: web controller (e.g. Spring MVC controller) to handle requests from the web.
- OrdersService: interface that defines the logic related to orders.
- OrdersServiceImpl: The implementation of the orders service.
- OrdersRepository: An interface that defines how we get access to persistent order information
- JdbcOrdersRepository: An implementation of the repository interface.

option could be Package by layer or feature or component, Ports and Adapters




### 11.5.1 Package by layer

Code is organized into layers according to its behavior. Each layer depends only on the next adjacent lower layer. E.g.: webcode, business logic, persistence

- **Pros:** simple, Tied to technical view, Commonly used
- **Cons:** As software grows, layers become “fat”, It is not tied to a the specific “business” view (the same for any domain). Dependencies are still possible between not adjacent layers

### 11.5.2 Package by feature

Vertical slicing by feature

- **Pros:** It well reflects the domain, Everything about a functionality is in one package
- **Cons:** Suboptimal in terms of separation of implementation details and business logic

### 11.5.3 Ports and Adapters

Domain-specific code (inner circle) is well separated from implementation details (outer circles). The latter depends on the former: the dependencies flow from outside toward inside.

### 11.5.4 Package by component

Service-centric view. Everything needed for orders is in the OrdersComponent. Separation of concerns is maintained inside each compone

## 11.6 Architectural styles

Reusable set of structural principles and high-level design decisions on:

- Component topology: by technical capabilities/by domain
- Physical architecture: monolithic/distributed
- Deployment: system's granularity and deployment frequency
- Communication style: synchronous/asynchronous, protocols/calls/queues
- Data topology: monolithic database/separate data
- Team topologies: remember Conway's law

**Monolithic** = single deployment unit of all code
**Distributed** = multiple deployment units connected through remote access protocols

| Monolithic | Distributed |
|---|---|
| Layered architecture | Service-based |
| Pipeline architecture | Event-driven |
| Modular monolith | Space-based |
| Microkernel | Service-oriented |
| | Microservices |

# Chapter 12

# Monolithic Architectural Styles

## 12.1 Architectural Quanta

The architectural quanta is the smallest part of the system that runs independently. An architectural quanta estabilishes the scope for a set of architectural caracteristics. It features:

- Independent deployment from other parts of the architecture
- High functional cohesion
- Low external implementation static coupling
- Synchronous communication with other quanta

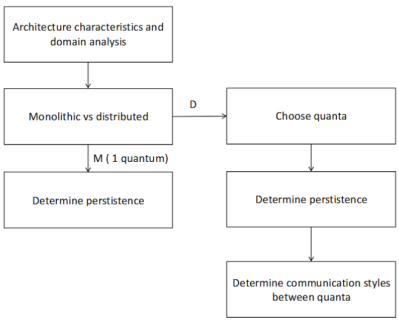



Figure 12.1: Scoping

## 12.2 Architectural styles

- Monolithic = single deployment unit of all code
- One architectural quantum
- Distributed = multiple deployment units connected through remote access protocols

| **Monolithic** | **Distributed** |
|---|---|
| Layered architecture | Service-based |
| Pipeline architecture | Event-driven |
| Modular monolithic style | Space-based |
| Micro-kernel architecture | Service-oriented |
| | Microservices |

## 12.3 Layered architecture - n tier

Also known as n-tier architecture. Technical partioning. Each layer is independent and has a specific responsibility. A layer represents a function of the system. Pro: testability

Common patterns are:

- Presentation layer
- Business logic layer
- Persistence layer
- Database layer

In the layered architecture, the layers can communicate only with the adjacent layers. Teams needing structured development.


### 12.3.1 Layered architecture - variations

### 12.3.2 Addding layers

New layers to enforce architectural constraints, helps controll access and dependencies. Example: add *service layer* for share common logic

### 12.3.3 Isolation

Each layer is indepedent, interacting only through defined contracts. Prevents tight coupling and cross-layer dependencies. The risk is the **Sinkhole antipattern**.

### 12.3.4 Open layers

Allow direct access to any layer for efficiency, useful for simple operations. Improves performance but reduces architectural control.

### 12.3.5 Open vs close layers

- Closed layers → stability, modularity, easier changes
- Open layers → flexibility, performance, but higher risk of coupling

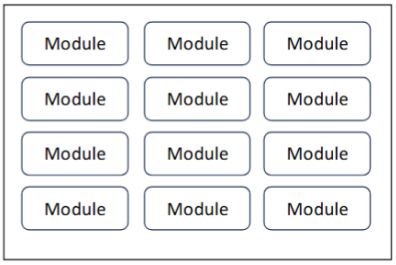
Synkholes acceptable threshold: 20% manageable, 80% change architecture style!

| | **Characteristic** | **Rating** | **Notes** |
|---|---|---|---|
| Structural | Simplicity | Very High | |
| | Modularity | Very low | |
| Engineering | Maintainability | Very low | Changes can impact every layer |
| | Testability | Low | Mock/stub components |
| | Deployability | Very low | At every change ! |
| | Evolvability | Very low | Changes can impact every layer |
| Operational | Responsiveness | Medium | Caching and multithreading |
| | Scalability | Very low | Lack of modularity |
| | Elasticity | Very low | Lack of modularity |
| | Fault tolerance | Very low | Faults impact the whole system, all layers must be restarted |

| **When to use** | **When not to use** |
|---|---|
| Applications with clear separation | Not ideal for high-performance or highly dynamic systems |
| Small, simple applications | Large and complex applications |
| As starting point for situations with low budget and short time | |
| To quickly test new ideas | |
| Examples of usage: operating systems, desktop applications | |

## 12.4 Modular monolithic style

Domain partitioned: each module encapsulates a business capability. Strong internal boundaries required. Modules with clear responsibilities. Deployed as a single unit of software. E.g. Web archive (WAR) file.7
Modules might have their own dedicated repositories.
In this we decupled all the modules like Customer, Order, etc. The team is also oragnized in different way before a team for each layer, now we have a team for each modules or a pool of modules.




#### Data topology

Also in this style we can have single database ormore domain-specific databases for the best performance.

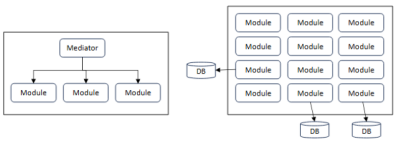
#### Interaction between modules

The modules **communicate directly** (peer to peer) or **through a mediator** the gool is orchestrate different modules.

### 12.4.1 Layered vs modular: components organization

- Layered: com.app.presentation.customer.profile
- Modular: com.app.customer.profile; com.app.customer.profile.presentation; ...

### 12.4.2 Variations of modular monolithic style

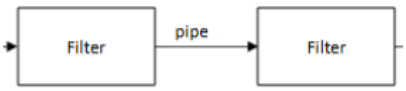



### 12.4.3 Advantages and disadvantages of modular monolithic style

| Key characteristics | Tradeoffs: |
|---|---|
| Simpler operational model than microservices | Risk of becoming tightly coupled over time |
| No network overhead | Limited independent scalability |
| Easier testing and debugging with respect to layered style | Large codebase can slow development |
| Requires discipline to maintain boundaries | Harder team autonomy at scale |
| | Risk of long start up |

## 12.5 Pipeline architecture

Also known as Pipes and Filters. Technically partitioned. Data flows through a sequence of processing steps. Each step transforms input into output. Filters are independent and stateless (ideally)

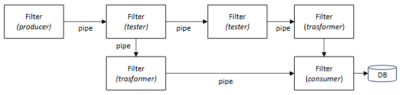


| | Characteristic | Modular Monolith | Notes |
|---|---|---|---|
| Structural | Simplicity | Very High | |
| | Modularity | Low | |
| Engineering | Maintainability | Low | Better than Layered because of higher modularity |
| | Testability | Low | As above |
| | Deployability | Low | As above |
| | Evolvability | Low | As above |
| Operational | Responsiveness | Medium | |
| | Scalability | Very low | |
| | Elasticity | Very low | |
| | Fault tolerance | Very low | |

| When to use | When not to use |
|---|---|
| Early-stage systems needing structure | Systems with high demands for operational characteristics |
| Situations with tight time and budget constraints | Systems requiring independent scaling |
| Teams wanting simplicity over distribution | When majority of changes are technically-oriented |
| Teams working with DDD | |
| Applications with strong domain boundaries | |

### 12.5.1 Components

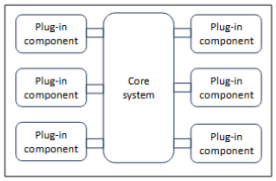
- **Filters**: processing units performing specific business functions
- **Pipes**: connectors passing data between filters, unidirectionally. Can be deployed as remote service (style remains monolith). Can be either synchronous or asynchronous (e.g., threads, messages).
- **Data format**: must be consistent or transformed
- **Database**: usually a single db, but filter can have their own database too
- **Producer**: starting point of a process
- **Transformer**: enhance, transform, compute (map)
- **Tester**: test input w.r.t. to criteria (reduce), optionally produces output
- **Consumer**: termination point, with persistence




| Key characteristics: | Tradeoffs: |
|---|---|
| Easy to extend by adding filters | Latency due to multiple stages |
| Clear separation of concerns | Harder debugging across pipeline |
| Examples: Bash, Zsh, MapReduce | Weak fit for complex business logic |
| | Data format coupling between filters |

## 12.6 Micro-kernel - plug-in architecture style

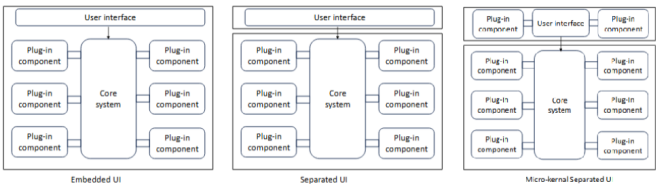
Also known as Plug-in architecture. Core system with basic and stable functionalities («happy paths»). It can be technically or domain partioned. Plug-ins extend system capabilities and incorporate complexity. Registry or discovery mechanism. Data topology: one db or plug-in specific data sources



| | Characteristic | Pipeline | Notes |
|---|---|---|---|
| Structural | Simplicity | High | |
| | Modularity | Low | Separation of concerns in filters |
| Engineering | Maintainability | Low | |
| | Testability | Medium | Separation of concerns in filters |
| | Deployability | Low | |
| | Evolvability | Medium | |
| Operational | Responsiveness | Medium | |
| | Scalability | Very low | Separate deployments can improve it, but at the expense of simplicity and cost. |
| | Elasticity | Very low | As above |
| | Fault tolerance | Very low | As above |

| When to use | When not to use |
|---|---|
| Data processing and transformation pipelines | In systems with non-deterministic flows (use event-driven a.s.) |
| Streaming or ETL systems | When fault tolerance is important |
| If properly implemented: independent, reusable processing steps | When bi-directional communication is needed |
| | If high demands of operational characteristics |

### 12.6.1 UI: variants




### 12.6.2 Plug-in components

- They should be independent and self-contained
- Communication p2p with the core system, not between plugins
- Compile based vs runtime base
- Common semantic, e.g. app.plugin.<domain>.<context> -> Example. app.plugin.editors.plain
- Plug-in could a standalone remote service
  - The architecture remains monolithic due to coupling with core system
  - Advantages: scalability, throughput, asynchronous communication
  - Disdvantages: reliability, costs, complexity

## 12.7 Monolithic Styles - recap

| **Characteristics:** | **Trade-offs:** |
|---|---|
| Highly extensible and customizable | Performance overhead from indirection |
| Separation between core and features | Complex plug-in lifecycle management |
| Supports product variability | Testing combinations of plugins can be hard |
| Stable core is critical | Core design mistakes are costly |

| **When to use** | **When not to use** |
|---|---|
| Common Product-based systems | Core systems that need frequent changes |
| Systems requiring extensibility | High dependencies between plugins |
| Platforms with plugin ecosystems | |
| Customization-driven applications | |
| Examples: browsers, IDEs, PMD, Jira | |

| | **Characteristic** | **Microkernel** | **Notes** |
|---|---|---|---|
| Structural | Simplicity | High | |
| | Modularity | Medium | Extensions easier with plugins |
| Engineering | Maintainability | Medium | Functionalities are better isolated w.r.t other monolithic styles |
| | Testability | Medium | As above |
| | Deployability | Medium | As above |
| | Evolvability | Medium | Extensions easier with plugins |
| Operational | Responsiveness | Medium | Microkernel applications are generally small |
| | Scalability | Very low | |
| | Elasticity | Very low | |
| | Fault tolerance | Very low | |

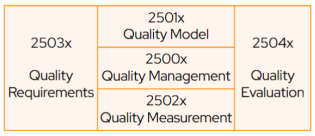
| | **Characteristic** | **Layered (n-tier)** | **Modular Monolith** | **Pipeline** | **Microkernel** |
|---|---|---|---|---|---|
| Structural | Simplicity | Very High | Very High | High | High |
| | Modularity | Very low | Low | Low | Medium |
| Engineering | Maintainability | Very low | Low | Low | Medium |
| | Testability | Low | Low | Medium | Medium |
| | Deployability | Very low | Low | Low | Medium |
| | Evolvability | Very low | Low | Medium | Medium |
| Operational | Responsiveness | Medium | Medium | Medium | Medium |
| | Scalability | Very low | Very low | Very low | Very low |
| | Elasticity | Very low | Very low | Very low | Very low |
| | Fault tolerance | Very low | Very low | Very low | Very low |

# Chapter 13

# Software Quality

ISO/IEC 9126: issued in 1991 and revised in 2001, being retired. Iso/IEC 25XX - SQuaRE: Software Product Quality Requirements and Evaluation. Is a family of standards in development.

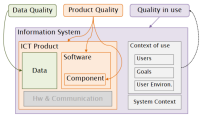
Software quality is divided in two parts: Product quality (internal, external, in use, data) and process quality.





## 13.1 Target entities

Product quality addresses several different entities in a complex information system




## 13.2 Model structure

- **Characteristics**: main aspects eg. usability
- **Sub-characteristics**: specific aspects eg. accessibility
- **Metrics**: evaluate a specific (sub)-characteristic, eg. number of clicks to reach a page
- **Measure elements**: Fundamental

## 13.3 Software product quality

- Functional suitability
- Performance efficiency
- Compatibility
- Interaction capability (Usability)
- Reliability
- Security
- Maintainability
- Flexibility

### 13.3.1 Functional suitability

Capability of a product to provide functions that meet stated and implied needs of intended users when it is used under specified conditions: completeness, correctness, appropriateness.

### 13.3.2 Performance efficiency

Capability of a product to perform its functions within specified parameters and be efficient in the use of resources under stated condition: time behaviour, resource utilization, capacity.

### 13.3.3 Compatibility

Capability of a product or its component to exchange information with other products, systems or components, and/ or perform its required functions, while sharing the same hardware or software environment: co-existence, interoperability.

### 13.3.4 Interaction capability (Usability)

Capability of a product to be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use: Recognizability, learnability, operability.

### 13.3.5 Reliability

Capability a product, to perform specified functions under specified conditions for a specified period of time: maturity, availability, fault tolerance, recoverability.

### 13.3.6 Security

Capability of a product to protect information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization, and to defend against attack patterns by malicious actors: confidentiality, integrity.

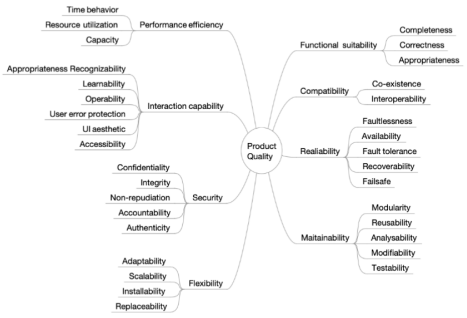
### 13.3.7 Maintainability

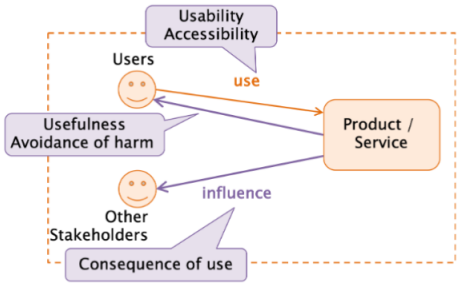
Capability of a product to be modified by the intended maintainers with effectiveness and efficiency: modularity, reusability, analysability, modifiability, testability.

### 13.3.8 Flexibility

Capability of a product to serve a different or expanded set of requirements or contexts of use; or the ease with which the product can adapt itself to changes in its requirements, contexts of use, or system environment: adaptability, installability, replaceability.

### 13.3.9 Sub-characteristics



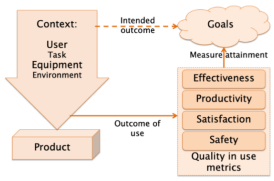

### 13.3.10 Usability - User error protection

| ID | UEp-1-G |
|---|---|
| Name | Avoidance of user operation error |
| Description | What portion of user actions and inputs are protected against causing any system malfunction? |
| Measurement function | X = A/B, where A = # user actions and inputs protected against causing any system malfunction; B = # user actions and inputs that could be protected |
| Interpretation | $0 \leq X \leq 1$ The closer to 1 the better |

### 13.3.11 Reliability - Maturity

| ID | RMa-1-G |
|---|---|
| Name | Test coverage |
| Description | How much of the required test cases has been executed during testing? |
| Measurement function | X = A/B, where A = # test cases actually performed representing operation scenario during testing; B = # test cases to be performed to cover requirements |
| Interpretation | $0 \leq X \leq 1$ The closer to 1 the better |

| ID | MMo-2-S |
|---|---|
| Name | Cyclomatic Complexity |
| Description | How many software modules have the acceptable cyclomatic complexity? |
| Measurement function | X = A/B, where A = # of software modules which have the cyclomatic complexity exceeds a threshold; B = # of software modules in a product. Threshold is defined by each project or organization and is possibly different value by a programming language, or type of module |
| Interpretation | $0 \leq X \leq 1$ The closer to 1 the better |




### 13.3.12 Maintainability - Modularity

## 13.4 Quality in Use

### 13.4.1 Quality in Use - Characteristics

- Effectiveness: accuracy and completeness with which identified goals are achieved
- Efficiency: resources used in relation to the results achieved
- Satisfaction: extent to which a user or other stakeholder perceives the results of use of a system, product or service to meet their needs and expectations
- Freedom from risk: degree to which the quality of a product or system mitigates or avoids potential risk to the user, organisation or project, including risks to economic status, human life, health, or the environment.
- Context coverage: degree to which a product or system can be used with effectiveness, efficiency, satisfaction, and freedom from risk in both specified contexts of use and in contexts beyond those initially explicitly identified.

### 13.4.2 Maintainability - Effectiveness

| ID | MMo-2-S |
|---|---|
| Purpose | Cyclomatic Complexity |
| Method of application | How many software modules have the acceptable cyclomatic complexity? |
| Definition | $M1 = \|1 - \sum A_i\|$, where $A_i$ = # Proportional value of each missing or incorrect component in the task output |
| Interpretation | $0 \leq M1 \leq 1$ The closer to 1 the better |

### 13.4.3 Quality in Use Influences

- Influence by use: usability and accessibility
- Influence by use of output: Perceived usefullness and Avoidance of harm
- Influence of context: Consequence of use
- Context coverage: Context completenss and Context extensibility

## 13.5 Data quality

Characteristics from two perspectives:

- **Inherent**: Focus on the data itself, their structure, schema, and semantics
- **System dependent**: Focus on how the data is stored and processed

| Inherent | Inher. / Sys. Dep. | Sys. Dep. |
|---|---|---|
| Accuracy | Accessibility | Availability |
| Completeness | Compliance | Portability |
| Consistency | Confidentiality | Recoverability |
| Currency | Efficiency | |
| Credibility | Precision | |
| | Understandability | |
| | Traceability | |

### 13.5.1 Accuracy

Correspondence between data and reality

- **Syntactic**: Value belongs to a set of validated information
- **Semantic**: The meaning, the content, corresponds to the reality

**Accuracy: Open vs. Closed world**

- Closed World Assumption (CWA): The knowledge represented in the data (and its schema) is complete. E.g., if a code appears in the list of valid codes it is accurate, otherwise it is wrong.
- Open World Assumption (OWA): The knowledge about the data is (knowingly) incomplete E.g., if a code is in the list of valid ones it is accurate, otherwise it is not possible to immediately decide

# Chapter 14

# Software Observability

**Observability:** is the ability to understand the internal state of a system by examining its outputs.

Telemetry data - i.e. logs and metrics allows:

- detecting possible problems
- troubleshooting and handle novel problems (unknown unknowns)
- understanding the motivation for system behavior (why is this happening?)

## 14.1 OpenTelemetry

OpenTelemetry is an observability framework and toolkit designed to facilitate the generation, export and collection of telemetry data such as traces, metrics, and logs. Open source, as well as vendor- and tool-agnostic.

### 14.1.1 Key concepts

- **Telemetry**: data emitted by a system during operation, requires instrumentation of the system
- **Service Level Indicator (SLI)**: measures the behavior of the system and en captures the user perspective
- **Service Level Objective (SLO)**: attaches business value to one or more SLIs

### 14.1.2 Distributed Tracing

Distributed tracing observe requests as they propagate through complex, distributed systems It is based on three main type of signals: **Logs**, **Spans** and **Traces**.

#### Logs

A log is a timestamped message emitted by services or other components. Built-in logging facilities in most languages, often lack contextual information.

#### Log format

- Structured: consistent schema or typed fields, can be JSON or other
- Unstructured: don't follow a consistent structure, therefore much more difficult to parse and analyze at scale
- Semistructured: include machine-readable key/value pairs or delimited fields but do not guarantee a stable schema

**Common Log Format (CLF)** 192.168.1.30 - alice [01/May/2025:07:20:10 +0000] "GET /index.html HTTP/1.1" 200 9481

### 14.1.3 Span

A span represents a single unit of work or operation. Spans describes specific individual operations triggere by a (user) request. It contains:

- name
- time-related data
- structured log messages
- other metadata (i.e. Attributes)

**Span Kind**

- Client: outgoing remote call such as an outgoing HTTP request or database call
- Server: incoming remote call such as an incoming HTTP request or remote procedure call
- Internal: operations which do not cross a process boundary
- Producer: creation of a job which may be asynchronously processed later
- Consumer: processing of a job created by a producer and may start long after the producer span has already ended

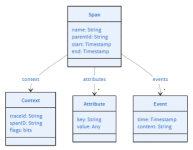
### 14.1.4 Trace

A (distributed) trace records the path taken by a single request as it propagates through multiple services/components.

A trace is made of one or more spans, the first span represents the *root span*. (span represents a request from start to finish). The spans underneath the parent provide a more in-depth context of what occurs.

## 14.2 Context Propagation

All spans part of the same trace must be identified an put in relation. Context information is passed to called components Standard traceparent HTTP header field: version, trace-id, parent-id, trace-flags




## 14.3 Metrics

A metric is a measurement of a service captured at runtime. Metric event: measure values, timestamps and other metadata (i.e. Attributes).

Capture a measurement: Name, Kind, Unit, Description.

### 14.3.1 Metric kind

- Counter: A value that accumulates over time
- UpDownCounter: A value that accumulates over time, but can also go down again, e.g. queue length
- Gauge: Measures a current value at the time it is read
- Histogram: A client-side aggregation of values, such as request latencies

Asynchronous: used if you access availablt only to the aggregated/current value

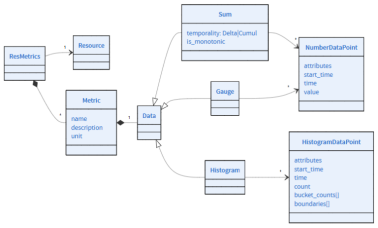
**Entities Example** App, browser, container, host, process, thread, service, etc.

**Metrics examples** For entity process:

| Name | Type | Unit | Description |
|---|---|---|---|
| cpu time | Counter | s | Total CPU seconds (by mode) |
| cpu utilization | Gauge | 1 | $\frac{\Delta time}{(elapsed * \#CPU)}$ |
| memory usage | UpDownCounter | by | Amount of physical memory in use |

For entity httṗserver:

| Name | Type | Unit | Description |
|---|---|---|---|
| request duration | Histogram | s | Duration of requests (attributes: method, url, ...) |
| active_requests | UpDownCounter | request | Number of active requests (attributes: method, url, ...) |
| request body size | Histogram | by | Payload size in bytes (attributes: method, url, ...) |




**Metric Example** **Model**

- Resource: namespace: my-site, name: frontend-web, id:http-server-136
- Metric: active_requests
- Data point: value: 35, method: GET, url: /index.html

## 14.4 Profiling

A running program consumes computational resources such as memory, CPU cycles, and elapsed time. To **profile** a program is to measure the consumption of such resources by specific elements of the program.

### 14.4.1 Why profiling?

Profiling means:

- make program more efficient
- developers more productive
- by identifying which program elements to optimize

The risk without profiling is to waste time in optimizing a method consuming few resources resulting in little impact.

### 14.4.2 Sampling vs. Tracing

Two main approaches to collect profiling data: **Sampling**, an external harness samples the status of the program and **Tracing**, an instrumented program emits samples during execution.

#### Sampling

Execution sampling providea an approximation of CPU profiling.

- Periodically interrupts the program (e.g., every 1ms)
- Records what the CPU is doing at that moment
- Low overhead, Works in production
- Not perfectly precise, but statistically accurate

#### Tracing

Tracing - a.k.a. instrumentation profiling - records function calls (all or selected ones).

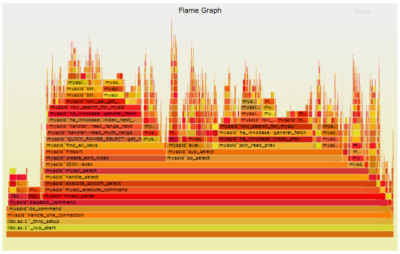
- High detail (exact timings, call counts)
- Higher overhead
- Can distort performance

### 14.4.3 Profile Types

- On-CPU
- Stall cycles: block on processor or hardware resources — typically memory I/O
- Memory: events related to memory growth e.g. malloc(), mmap(), etc.
- I/O: issuing of I/O, such as file system, storage device, and network
- Off-CPU: time spent while a thread is blocked, not running on a CPU (off-CPU time), e.g., waiting on I/O, locks, timers, a turn on-CPU, waiting for paging or swapping.

### 14.4.4 Flamegraph

Collection of stack samples represented as a column of boxes, each box represents a function (a stack frame).
The y-axis shows the stack depth, from root at the bottom to leaf at the top. The top box is the function that was on-CPU
The x-axis spans the stack trace collection (not time passing). Ordering is alphabetical on the function names. When identical boxes are adjacent, they are merged.
The width of box is the frequency at which that function was present in samples. Functions with wide boxes were more present in the stack traces than those with narrow boxes, in proportion to their widths.
The background color is random to help the eye differentiate boxes.




captionoffigureExample of flamegraph

**Flamegraph interpretation**

The top is the function that was on-CPU when the stack trace was collected. Look for large plateaus along the top edge: this means a single function was frequently running on-CPU. Reading top down shows ancestry. Reading bottom up shows code flow and the bigger picture

The width of function boxes can be directly compared, if a function box is wider than another, this may be because it consumes more CPU per function call or that the function was simply called more often.

Major forks in the flame graph can indicate, a logical grouping of code (work in stages), a conditional statement.

## 14.5 Java Flight Recorder (JFR)

JFR is a low-overhead profiling and monitoring system built into the JVM. It records events about CPU, memory, GC, threads, I/O, and JVM internals. Data is stored in a compact .jfr file for later analysis. It supports both: continuous production monitoring and short profiling sessions.

### 14.5.1 Collected Events

- JVM Internals (e.g. CPU, Threads, Monitors)
- Profiling (execution sampling)
- Garbage Collection
- Memory (object allocation)
- Exception
- I/O
- Class loading

### 14.5.2 JFT settings

Two predefined settings:

- default: No jdk.ExecutionSample; Focus on: GC, locks, I/O, JVM internals; Very low overhead
- profile: Enables: jdk.ExecutionSample + allocation profiling; Higher overhead ( 1-2%); Suitable for flamegraphs

**JFR profiling session example** An app is run with JFR started from begin:

```
 Event Type          Count Size (bytes)
=========================================
 jdk.NativeLibrary   1516  129323
 jdk.ModuleExport    1017   10447
 jdk.SystemProcess   1002  104888
 jdk.BooleanFlag      966   28708
 jdk.ActiveSetting    752   18683
 jdk.GCPhaseParallel  319    7794
 jdk.ModuleRequire    312    2964
```

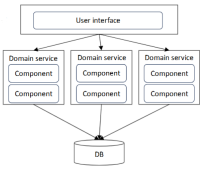
```
java \
    -XX:StartFlightRecording=filename=profiling-recording.jfr
    -cp target/classes \
    it.polito.swda.profiling.Example

jfr summary profiling-recording.jfr
```

# Chapter 15

# Distributed Architectural Styles

### 15.0.1 Service-based Architecture

Distributed macro-layered structure: UI+ services
Quanta: UI separately deployed. Separately deployed coarse-grained services.
Optionally, monolithic database
Domain services are independent, usually do not communicate each other




### 15.0.2 Domain services design

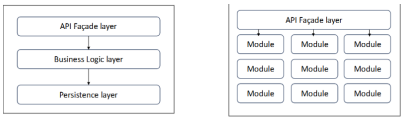



Figure 15.1: Technical partitioning and Domain partitioning

### 15.0.3 API access, UI and DB options

API access façade acts as orchestrator: failures can be handled at this level.
UI/DB options: Single monolithic, Domain-based or Service-based.
API gateway options between UI and domain services.

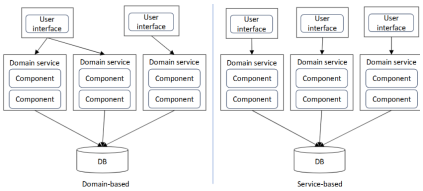



Figure 15.2: UI options

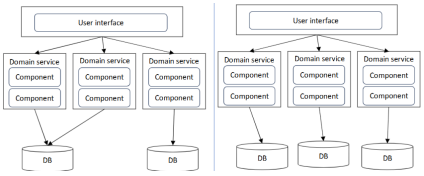
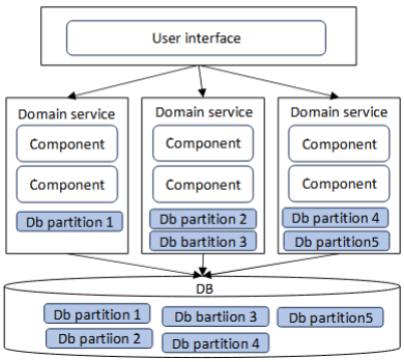



Figure 15.3: DB options

### 15.0.4 Database logical partitioning

Using a single shared library for DB entity objects impacts all services in cases of changes. Logical partitioning and multiple share libraries for db entity objects reduces coupling. It should be as fine-grained as possible

Using a single shared library for DB entity objects impacts all services in cases of changes. Logical partitioning and multiple share libraries for db entity objects reduces coupling. It should be as fine-grained as possible.




**When to Use**

- Small-medium systems
- Limited scalability needs
- No or low need of service coordination
- Teams new to distributed systems
- As intermediate steps towards migration to other distributed styles

### 15.0.5 Service-based architecture: characteristics

| | Characteristic | Rating | Notes |
|---|---|---|---|
| Structural | Simplicity | Medium | |
| | Modularity | Medium | Domain scoping, federated UI and BD possible |
| Engineering | Maintainability | High | As consequence of modularity |
| | Testability | High | As above |
| | Deployability | High | As above |
| | Evolvability | High | As above |
| Operational | Responsiveness | Medium | |
| | Scalability | Medium | Coarse-grained services |
| | Elasticity | Low | |
| | Fault tolerance | Medium | Self-contained services |

## 15.1 Microservices architecture

Domain partitioning took to the extreme. Heavily inspired by the idea of domain-driven-design and DevOps
Fundamental concept of "bounded context"

- Everything related to a portion of the domain is visible internally but opaque to other bounded contexts
- It drives systematically interventions of decoupling

Protocol-aware heterogenous interoperability

- Standardized communications (e.g., REST)
- Independent services
- Inter-service communication possible

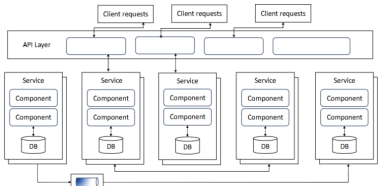



Figure 15.4: Overview

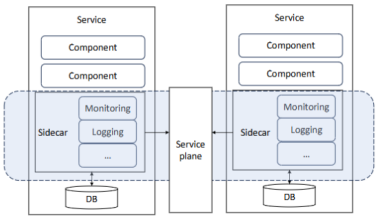



Figure 15.5: Monitoring and governance: sidecar pattern

### 15.1.1 Granularity of services

Extreme granularity may affect performances
Main decision criteria for micro-services:

- Purpose (i.e., problem domain)
- (Avoid/minimize) transactions
- Choreography

Other aspects:

- Data isolation: per service or per domain
- API layer "neutrality"
- Frontends: monolithic vs coupling with backend services
- Choreography vs orchestration

| | Characteristic | Rating | Notes |
|---|---|---|---|
| Structural | Simplicity | Very low | |
| | Modularity | Very high | |
| Engineering | Maintainability | Very high | |
| | Testability | Very high | |
| | Deployability | Very high | |
| | Evolvability | Very high | |
| Operational | Responsiveness | Low | Many network calls to complete work; data caching and replication + choreography can improve it |
| | Scalability | Very high | |
| | Elasticity | High | |
| | Fault tolerance | Very high | |

### 15.1.2 Microservices architecture: characteristics

## 15.2 Event-driven architecture

Main implementation: choreographed and technically partitioned Decoupled event processing components asynchronously trigger and responds via events. Asynchronous fire-and-forget communication. Loose coupling.

Four primary components:

- Initiating event
- Event broker
- Event processor
- Derived events

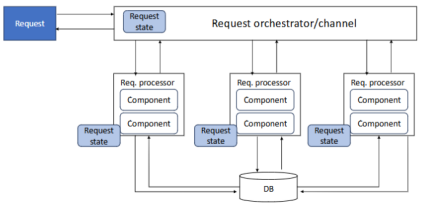



Figure 15.6: Event-driven architecture

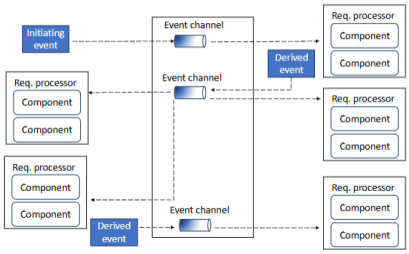
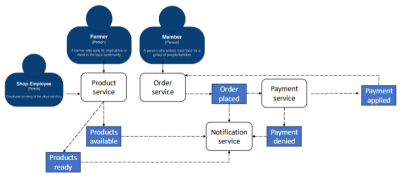



Figure 15.7: Fire and forget




**EDA processing - example with SPG**

### 15.2.1 Events vs messages

### 15.2.2 Further options

- **Triggering extensible events**: It increases evolvability of the systems
- **Asynchronous vs synchronous = performance vs responsiveness**: NB asynchronous communication decouples architectural quanta
- Broadcast capabilities for implementing semantic decoupling
- Granularity of events: Be aware of swarm of gnats antipattern !
- Error handling through Workflow Event pattern or tools

| | EVENTS | MESSAGES |
|---|---|---|
| **ACTION** | Reacting to something already happened | Obeying to a request with something that needs to be done |
| **ANSWER** | It doesn't require an answer | It -typically- requires an answer |
| **COMMUNICATION STYLE** | One-to-many (e.g., publish and subscribe) | One-to-one (or point-to-point) |
| **COMMUNICATION CHANNEL** | Topic, stream, notification service | Queue, messaging service |

### 15.2.3 Event payload

**Data-based payload**

- It avoids pression on DB: payload contains all information for subscribers
- Extra support for data consistency and integrity
- Data format creates coupling: Stamp coupling
- Versioning: Data and Contract

**Key-based payload**

- Payload contains pairs of keyvalues to later query a DB. Typically, JSON or XML, Human readable
- DB remains the single source of information
- Minimum contracts and network usage
- No need of versioning

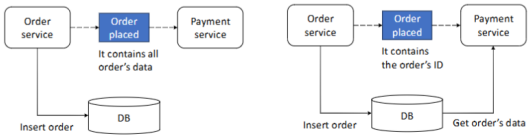



Figure 15.8: Data-based payload, Key-based payload

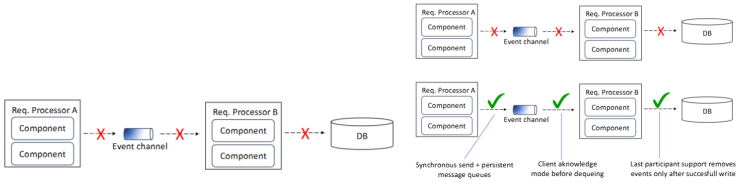
**Event payload - Example**

### 15.2.4 Events payload: tradeoffs

| | DATA-BASED | KEY-BASED |
|---|---|---|
| **Performance and scalability** | ☺ | ☹ |
| **Contract management** | ☹ | ☺ |
| **Stamp coupling** | ☹ | ☺ |
| **Bandwidth utilization** | ☹ | ☺ |
| **Restricted db access** | ☺ | ☹ |
| **Overall system fragility** | ☹ | ☺ |

### 15.2.5 Data loss prevention

1. While A publishes an event, it crashes before ack from event broker
2. B accepts an event, but it crashes before processing it
3. A data error prevents B from ensuring persistence



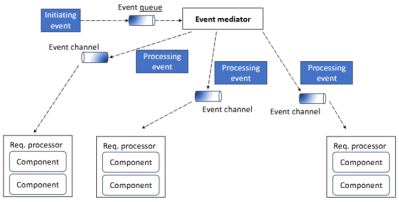
### 15.2.6 Request-reply processing

Also known as *pseudosynchronous communications*. It requires a request queue and a reply queue. Event producers waits until confirmation of event producer is retrieved fomr the reply queue.

Two common implementations:

- Correlation ID
- Temporary queue with specific id

## 15.3 Orchestrated EDA




Orchestration vs Choreography.
Mediation by a specific component/service.
Event mediator typically uses messages. The mediator is responsible of catching events and generate specific point-to-point derivated events. Usually with queues.
Without further communicating its own actions (fire and forget principle)
Orhestrator's options: monolithic, domain-based, service-dedicated.

### 15.3.1 Event drive architecture: usage

**When to use**

- Complex workflows, with thousands scenarios
- Dynamic user content
- Reactive systems
- Real time processing

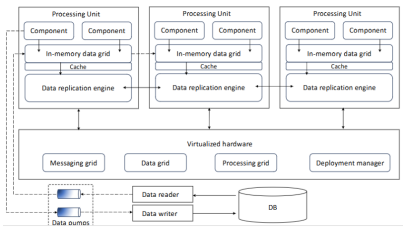
**When NOT to use**

- High needs of synchronous communication
- Static coupling (contracts) increases
- Dynamic coupling (sync calls) increases

### 15.3.2 EDA architecture: characteristics

| | Characteristic | Rating | Notes |
|---|---|---|---|
| Structural | Simplicity | Low | Dependent also on dynamic coupling and data topologies |
| | Modularity | High | |
| Engineering | Maintainability | High | |
| | Testability | Low | Due to non-deterministic workflow |
| | Deployability | Medium | |
| | Evolvability | Very high | |
| Operational | Responsiveness | Very high | |
| | Scalability | High | Additional processors can be added; db still bottleneck |
| | Elasticity | Medium | |
| | Fault tolerance | Very high | |

## 15.4 Space-based architecture




Technically partitioned. Complex implementation. Designed to optimize operations with large volumes of data, users, and load peaks.
Main features:

- Eliminates database bottleneck
- Distributed data management
- Distributed processing units

### 15.4.1 Space-based architecture: elements

### 15.4.2 Replicated cache data: collisions

Collisions typically occur when the update rate of the data in the cache exceeeds the replication latency.

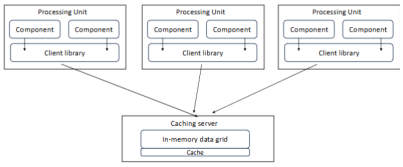
Several factors influence the amount of data collisions:

| Element name | Description |
|---|---|
| Processing units | Contain the application functionality |
| Virtualized middleware | The collection of infrastructure-related artifacts that are used to manage and coordinate the processing units |
| Messaging grid | It manages input requests and session states (e.g., a web server with load balancing) |
| Data grid | Manages the synchronization and replication of data between processing units. It can be implemented also in processing units. |
| Processing grid | Used to manage request orchestration between multiple processing units; optional |
| Deployment manager | Manages the starting and tearing down of processing unit instances based on load |
| Data pumps | Asynchronously send updated data to the database. It decouples processing units and database. Usually implemented with messaging and queues. Can be mapped to domains |
| Data writers | Perform the updates from the data pumps. They can be multiple and dedicated to data pumps |
| Data readers | Read database data and deliver it to processing units (upon startup) through reverse data pumps. |

- N : # of processing unit instances containing the same cachce
- UR: cache update rate (updates/seconds)
- S: cache size (# of rows)
- RL: Replication latency (milliseconds)

$$\text{Collision Rate} = N \times \frac{UR^2}{S} \times RL$$

### 15.4.3 Distributed caching




### 15.4.4 Replicated vs distributed caching

= Asynchronous replication vs synchronous replication Replicated caching:

- requires in-memory data grid, very fast
- difficult with very high data volumes and frequent updates
- possible also wih near-cache techniques: a caching server + partial front caches

| Criteria | Replicated | Distributed |
|---|---|---|
| Optimization | Performance | Consistency |
| Cache size | Small | Large |
| Type of data | Mostly static | Highly dynamic |
| Update frequency | Low | High |
| Fault tolerance | High | Low |

## 15.5 SBA architecture - Usage

### 15.5.1 When to use:

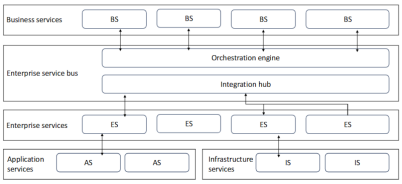
- High volume applications
- Large, concurrent user load
- Low-latency systems

### 15.5.2 Space based architecture: characteristics

| | Characteristic | Rating | Notes |
|---|---|---|---|
| Structural | Simplicity | Very low | |
| | Modularity | Medium | Technically partitioned |
| Engineering | Maintainability | Medium | |
| | Testability | Very low | Difficult to simulate very high loads |
| | Deployability | Medium | |
| | Evolvability | Medium | |
| Operational | Responsiveness | Very high | |
| | Scalability | Very high | |
| | Elasticity | Very high | |
| | Fault tolerance | Low | Technically partitioned, distributed computing and data management |

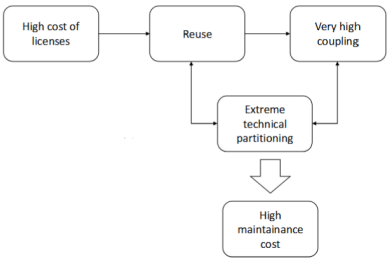
## 15.6 Orchestration-driven service-oriented architecture

- Enterprise-level architecture
- Layers with specific responsibilities
- Strong governance through taxonomy of services
- Services share enterprise bus
- Usage in companiesdecreasing, due to extreme technical partitioning



### 15.6.1 Service-oriented architecture: layers

- **Business services:** no business logic, no code, only I/O schemas (service signatures)
- **Entrerprise service bus:** Transaction coordination and message transformation. Also for internal calls, it can "eat" the whole architecture
- **Enterprese services:** Building blocks of isolated and combinable/reusable functionalities
- **Application services:** one-off, single implementation services, typically not reusable
- **Infrastracture services:** Monitoring, authentication, logging




**Reasons for decay after the 2010s** Domains concepts spread so thinely that they were virtually ground to dust

### 15.6.2 ODSA architecture: characteristics

| | Characteristic | Rating | Notes |
|---|---|---|---|
| Structural | Simplicity | Very low | |
| | Modularity | High | |
| Engineering | Maintainability | Very low | Because of very high coupling |
| | Testability | Very low | As above |
| | Deployability | Very low | |
| | Evolvability | Very low | Ripple effects for any functional/domain change |
| Operational | Responsiveness | LOW | |
| | Scalability | High | Through session duplication |
| | Elasticity | Medium | |
| | Fault tolerance | Medium | |

## 15.7 Distributed Styles Architectural Characteristics: synopsis

### 15.7.1 Structural and engineering characteristics

| | Characteristic | Service based | Microservice | Event driven | Space based | Orch. Driven Service or. |
|---|---|---|---|---|---|---|
| Structural | Simplicity | Very low | Very low | Low | Very low | Very low |
| | Modularity | Very high | Very high | High | Medium | High |
| Engineering | Maintainability | Very high | Very high | High | Medium | Very low |
| | Testability | Very high | Very high | Low | Very low | Very low |
| | Deployability | Very high | Very high | Medium | Medium | Very low |
| | Evolvability | Very high | Very high | Very high | Medium | Very low |

### 15.7.2 Operational characteristics

| | Characteristic | Service based | Microservice | Event driven | Space based | Orch. Driven Service or. |
|---|---|---|---|---|---|---|
| Operational | Responsiveness | Medium | Low | Very high | Very high | Low |
| | Scalability | Medium | Very high | High | Very high | High |
| | Elasticity | Low | High | Medium | Very high | Medium |
| | Fault tolerance | Medium | Very high | Very high | Low | Medium |

# Chapter 16

# Deployment

## 16.1 Deployment strategies

### 16.1.1 Infrastructure deployment

- On premise: Software system installed and run on servers physically located in the organization
- Hosted: Software system installed on third parties centers
- Cloud: Software system is virtualized over the cloud

### 16.1.2 Application deployment strategies

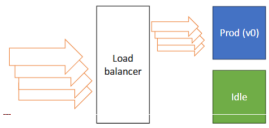
**Goal**: define how an application, or new versions of it, are deployed on an infrastructure
**Main strategies**: In place upgrade, Blue-Green, Canary / Rolling

#### In place upgrade

Installation on top of current version. It requires downtime. All existing settings are preserved

#### Blue-green

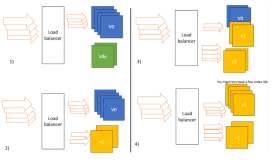
Two environments, a load balancer directs traffic. Blue env.: where the current application runs. Green: idle
When a new version is ready for deployment:




- Update deployment, test and monitoring on the green environment
- When situation is deemed stable, all traffic is sent to the green env.
- The green environment now is the main one, the blue is idle

### 16.1.3 Canary

"Canary in the coal mine".
Initially, only a few nodes hosts the new deployed application version. Load balancer forwards a minority fraction of requests to new version. Performances are monitored. If there are no severe issues:



- load balancer increases the percentage towards new application version;
- The new application is deployed in further nodes

This incremental process continues until the new version is handling all traffic on the majority of nodes, and few ones remains idle, so that process can restart with further new versions.
In case of problems, balance can be temporary reversed

**Canary with feature flags**

- New features are flagged in the code, so that they can be enabled via configuration files
- New features will be enabled for a small groups of users
- Monitor the performances of the “canary” group
- If no significant issues arise, increase dimension of “canary” group
- Progressively all users will use the new features
- In case of problems, dimensions of canary group can be temporary diminished

**Rolling with feature flags**

- Same as canary, but exposure of new changes is made on nodes, not users
- New features are flagged in the code, so that they can be enabled via configuration files
- New features will be enabled for a small groups of (users) nodes
- Monitor the performances of the “canary” group
- If no significant issues arise, increase dimension of “canary” group
- Progressively all (users) nodes will use the new features
- In case of problems, dimensions of canary group can be temporary diminished

## 16.2 Deployment visualization

**Deployment diagram**

**Goal**: describe the mapping of software systems and containers (and any other deployable unit) onto deployment nodes
**Method**: C4 model with further boundaries for each deployment node
Deployment node: a device or execution environment

- Physical infrastructure (e.g. a physical server or device)
- Virtualised infrastructure (e.g. a virtual machine, IaaS)
- Containerised infrastructure (e.g. a Docker container)
- An execution environment (e.g. a database server)